CLINICAL PROTOCOL

IND No. 167326

An open-label, single ascending dose study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of intrathecally administered PrP-siRNA in healthy adult patients at genetic risk for prion disease.

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SHORT PROTOCOL TITLE

Safety and pharmacology of PrP-siRNA in pre-symptomatic prion disease

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Study Intervention Provided by:

Broad Institute of MIT and Harvard

Sponsor of IND/IDE: Broad Institute of MIT and Harvard
Sponsor representative: Eric Vallabh Minikel, PhD

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INVESTIGATOR AGREEMENT

I have read the foregoing protocol *[INSERT VERSION NUMBER AND DATE]* and agree to conduct the study as described herein.

By signing the protocol, the Investigator agrees to keep all information provided by the Sponsor in strict confidence and to request the same from his/her staff and the Institutional Review Board. Study documents provided by the Sponsor will be stored appropriately to ensure their confidentiality. The Investigator should not disclose such information to others without authorization, except to the extent necessary to conduct the study.

_____ _____

Investigator Signature **Date**

_____

Print Investigator's Name

\

SIGNATURE PAGE

Study Number: TBD

Principal Investigator Approval:

Signature: _____ Date: _____

Name: *Insert PPI Name*

STATEMENT OF COMPLIANCE

The trial will be carried out in accordance with International Conference on Harmonisation Good Clinical Practice (ICH GCP) and the United States (US) Code of Federal Regulations (CFR) applicable to clinical studies (45 CFR Part 46, 21 CFR Part 50, 21 CFR Part 56, 21 CFR Part 312, and/or 21 CFR Part 812).

ABBREVIATIONS

AE	Adverse Event
ASO	Antisense oligonucleotide
CFR	Code of Federal Regulations
CJD	Creutzfeldt-Jakob disease
CRF	Case report form
CS	Clinically Significant
CSF	Cerebrospinal fluid
di-siRNA	Divalent short interfering RNA
dpi	Days post-infection
ED_{50}	Median effective dose
ELISA	Enzyme-linked immunosorbent assay
exNA	Extended Nucleic Acid
FDA	Food and Drug Administration
FFI	Fatal familial insomnia
GCP	Good Clinical Practice
GSS	Gerstmann-Straussler-Scheinker disease
HIPAA	Health Insurance Portability and Accountability Act
ICH	International Conference on Harmonization
IMM	Independent Medical Monitor
MedDRA	Medical Dictionary for Regulatory Activities
MRC	Medical Research Council
MRC-PDRS	Medical Research Council Prion Disease Rating Scale
PD	Pharmacodynamic
PK	Pharmacokinetic
PPI	Protocol Principal Investigator
PrP	prion protein
SAE	Serious adverse event
siRNA	short interfering RNA
SOA	Schedule of Activities
VPSPr	Variably protease-sensitive prionopathy

SYNOPSIS

Prion disease is a fatal and incurable neurodegenerative disease caused by prion protein (PrP)[1]. PrP is encoded by a chromosomal gene, *PRNP*, and is expressed in the brains of all humans and all mammals. In prion disease, PrP undergoes post-translational misfolding into a self-replicating conformer known as a prion. Prions spread exponentially throughout the brain by recruiting other molecules of PrP, eventually accumulating to a level where neurotoxicity ensues. Subtypes of prion disease known historically by various names including CJD, FFI, and GSS all share this fundamental molecular mechanism. Prion disease typically presents clinically as a rapidly progressive dementia. 85% of cases are sporadic, apparently due to stochastic protein misfolding events. 15% are genetic, due to gain-of-function mutations in *PRNP*[2], and <1% of cases are acquired through exposure to exogenous prions via dietary or iatrogenic routes[3]. Regardless of etiology, once symptomatic, patients decline rapidly. Overall median survival is 5 months from first symptom[4], but often 3 months are spent in diagnostic odyssey[5]; the median survival from enrollment in previous trials was approximately 2 months[6,7]. Prion disease has no effective therapy and no standard of care.

Extensive preclinical studies demonstrate that lowering PrP both delays the onset and slows the progression of prion disease in diverse animal models[8–16]. While PrP lowering is effective across different disease stages, it is most effective at pre-symptomatic stages. In pre-symptomatic mice, chronic ASO treatment lowering PrP by 50% tripled survival time, and this arose mostly from an increase in healthy lifespan as quantified by weight gain, nest-building, and observation of symptoms. In near-symptomatic or symptomatic mice, the overall survival benefit from ASO treatment was more modest (20% to 70% increase in survival time depending on the treatment timepoint and disease endpoint), and was derived mostly or entirely from slowing of symptomatic progression. Reversal of symptoms already accrued prior to dosing was <u>not</u> observed. Therefore, while PrP lowering is expected to be effective even in symptomatic patients, the greatest good that PrP lowering can do is to delay onset in pre-symptomatic at-risk individuals who still have full quality of life.

The therapeutic hypothesis of PrP lowering has only just yielded its first clinical test, as an antisense oligonucleotide (ASO) to lower PrP entered a Phase I/IIa clinical trial in symptomatic patients in January 2024. Results of this trial are not yet known. This ASO has not yet been administered to pre-symptomatic patients, and the sponsor (Ionis Pharmaceuticals) has not announced any intention or timeline for doing so.

The primary objective of this trial is to evaluate safety and tolerability of investigational drug PrP-siRNA (referred to elsewhere[15] as 2439-s4 or 2439-exNA) in pre-symptomatic subjects at genetic risk for prion disease. Secondary objectives are to evaluate the pharmacokinetics and pharmacodynamic impact of PrP-siRNA in this population. In rats[17] and macaques[18], CSF PrP responds to brain PrP lowering, suggesting that measurement of PrP in the CSF may serve as an adequate surrogate of brain target engagement for PrP-siRNA. Exploratory endpoints will include measures of disease progression and biomarkers of neurodegeneration.

<u>Study Title</u>

Full title: An open-label, single ascending dose study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of intrathecally administered PrP-siRNA in healthy adult patients at genetic risk for prion disease.

Short title: Safety and pharmacology of PrP-siRNA in pre-symptomatic prion disease

Objectives

The objectives of this study are to evaluate the safety, tolerability, pharmacodynamics, and pharmacokinetics of intrathecally administered PrP-siRNA in pre-symptomatic subjects at genetic risk for prion disease.

The objectives and endpoints are summarized in Table 1.

Table 1. Summary overview of objectives and endpoints for this clinical trial.

Ranking	**Objective**	**Endpoint**
Primary	To characterize the safety and tolerability of PrP-siRNA	Frequency of adverse events. Safety will also be evaluated through vital signs, physical exams, cognitive assessments, suicide severity assessment and clinical laboratory assessments.
Secondary	To characterize the pharmacokinetics of single dose PrP-siRNA	siRNA concentration in CSF and blood measured by mass spectrometry
	To characterize the pharmacodynamic effects of single doses of PrP-siRNA	Change from baseline CSF PrP concentration measured by ELISA
Exploratory	To characterize clinical onset or markers of progression towards onset	MRC-PDRS Brain magnetic resonance imaging Disease-related biomarkers in blood and CSF

Study Design

This is a first-in-human, open-label single ascending dose study in pre-symptomatic adults at genetic risk for prion disease. The study will consist of a screening period of 1-2 weeks, administration of a single drug dose, and a 24-week post-dose monitoring period. This study will enroll up to 15 participants.

Three dose level cohorts will be enrolled sequentially, with each participant receiving a single dose of either 50, 100, or 200 mg, for 3, 3, and 9 patients respectively. Sentinel dosing will be employed with each new dose escalation with a minimum of 14 days of post-dose safety prior to further dosing.

For each dose escalation, the decision to proceed to dosing the next cohort will be made by the Independent Medical Monitor (IMM), upon review of a minimum of 14 days of available post-dose safety data from the 2nd and 3rd patient of a dose cohort and minimum of 28 days of available post-dose safety data from the 1st patient of a dose group. If no stopping criteria have been reached, dose escalation is permitted.

Interventions and Duration

The investigational drug, named PrP-siRNA, is a divalent siRNA sequence-targeted to bind and degrade the RNA molecule encoding prion protein (PrP), the protein that causes prion disease. Dosing of this siRNA is hypothesized to lower PrP concentration in the brain and CSF. The product is administered intrathecally.

After informed consent is obtained, participants will be screened between Day -14 and Day 0. Eligible patients will be admitted to the clinical trial center on Day 0 to confirm continued eligibility and for pre-dose assessments including CSF collection. Patients will then receive a single dose of PrP-siRNA via intrathecal injection.

Assessments during the initial admission period include vital signs, physical exam, ECG monitoring, clinical labs, CSF labs and cognitive evaluation. Full neurological examination will be performed post-dose and prior to discharge from the clinical study center.

Participants will be discharged after 24 hours and then periodically return to the study center on an outpatient basis at Week 4, 8, 12 and 24 for safety monitoring and study activities through the 24 week follow up period. Each outpatient study visit will include lumbar puncture, phlebotomy, vital signs, physical exam, neurological exam, adverse event review, suicide severity rating scale, cognitive evaluation, clinical safety labs and CSF safety, PK and PD labs.

Sample Size and Population

This study will enroll up to 15 participants.

Key inclusion criteria include lack of clinically manifested symptoms of prion disease, in the opinion of the investigator; and a *PRNP* genetic test indicating the presence of a highly penetrant genetic variant causal for prion disease. Key exclusion criteria include pregnancy; contraindication to lumbar puncture; or recent participation in a different prion disease clinical trial. Full inclusion and exclusion criteria are provided in sections 4.1 and 4.2.

1 STUDY OBJECTIVES

The study objectives are summarized Table 1 above and are detailed below.

1.1 Primary Objectives

The primary objective is to assess the safety and tolerability of PrP-siRNA in pre-symptomatic adults at genetic risk for prion disease.

Primary endpoints serving this objective will include:

- Frequency of adverse events. Definitions and assessment procedures are provided in Section 7 below.
- Safety and tolerability will also be evaluated through vital signs, physical exams, cognitive assessments, suicide severity assessment and clinical laboratory assessments.

1.2 Secondary Objectives

The secondary objective is to assess the pharmacokinetics and pharmacodynamic impact of PrP-siRNA in symptomatic patients with prion disease.

Secondary endpoints serving this objective will include:

- Measurement of PrP-siRNA concentration by mass spectrometry in blood and CSF. This will permit us to characterize drug exposure in patients
- Measurement of PrP concentration in CSF by ELISA[17]. Each participant's CSF PrP concentration will be compared to their own individual baseline prior to investigational drug administration. PrP will be measured at 4, 8, 12, and 24 weeks post-dose, spanning the times when it is expected, based on nonclinical studies in mice, that PrP-siRNA target engagement may be maximal (4 weeks post-dose) and when PrP will have returned to within 20% of baseline (24 weeks post-dose).

1.3 Exploratory Objectives

The exploratory objective is to characterize disease onset or progression towards disease based on imaging and biochemical markers in this pre-symptomatic population.

Exploratory endpoints serving this objective will include:

- MRC-PDRS, a clinical measure of disease progression capturing ability to perform daily activities of life.
- Brain MRI to examine imaging signatures of prion disease and brain volume changes.
- Fluid biomarkers of prion seeding (CSF RT-QuIC), neurodegeneration (CSF NfL, CSF T-tau, CSF beta-synuclein, plasma brain-derived tau, plasma NfL), and neuroinflammation (plasma GFAP).

2 BACKGROUND

2.1 Rationale

Debilitating and rapidly fatal, prion disease has a profound unmet medical need. Prion disease is a fatal and incurable neurodegenerative disease caused by prion protein (PrP)[1]. PrP is encoded by a chromosomal gene, *PRNP*, and is expressed in the brains of all humans and all mammals. In prion disease, PrP undergoes post-translational misfolding into a self-replicating conformer known as a prion. Prions spread exponentially throughout the brain by recruiting other molecules of PrP, eventually accumulating to a level where neurotoxicity ensues. Subtypes of prion disease known historically by various names including CJD, FFI, and GSS all share this fundamental molecular mechanism. Prion disease typically presents clinically as a rapidly progressive dementia. 85% of cases are sporadic, apparently due to stochastic protein misfolding events. 15% are genetic, due to gain-of-function mutations in *PRNP*[2], and <1% of cases are acquired through exposure to exogenous prions via dietary or iatrogenic routes[3]. Regardless of etiology, once symptomatic, patients decline rapidly, with overall median survival of 5 months from first symptom[4]. At-risk individuals may know their genetic status decades in advance of symptom onset, yet at present there are no medications nor lifestyle measures known to delay the onset of the disease. ION717, an antisense oligonucleotide (ASO) designed to lower PrP, entered a Phase I/IIa clinical trial in January 2024, which is scheduled to read out in mid-2025, but the sponsor of that trial (Ionis Pharmaceuticals) has not dosed pre-symptomatic subjects nor announced a timeline for doing so. There remains an urgent need to develop additional modalities to delay or prevent prion disease.

***PRNP* genotype can identify individuals very likely to develop prion disease, before any pathological changes occur.** The target population for this trial is pre-symptomatic adults at high genetic risk for prion disease. All genetic cases of prion disease are caused by protein-altering germline DNA varaints in the *PRNP* gene, encoding PrP. Although mutations that confer only low lifetime risk have been identified, a subset of mutations are known to confer a very high (estimated >90%) lifetime risk of developing prion disease[2,19]. The high penetrance of these genetic variants has been demonstrated by their segregation with disease in multiple multigenerational families, by their observation as *de novo* variants in certain patients, by survival curves constructed from over one thousand patients, and by their dramatic enrichment in prion disease cases relative to their extreme rarity in the general population[2,19,20]. This protocol is focused on individuals with these high risk mutations. This includes any subtype of prion disease, regardless of classification as CJD, FFI, GSS, or other. Cross-sectionally, individuals with these high-risk genotypes exhibit normal biomarkers for most of their lives: their CSF RT-QuIC is generally negative, and their concentrations of CSF and plasma NfL and T-tau and other markers of neurodegeneration are indistinguishable from mutation-negative individuals[21,22]. When at-risk individuals do eventually convert to active disease, their RT-QuIC may become positive 1-3 years before symptom onset[22], but even in this period, CSF PrP levels remain unchanged, and any increase in markers of neurodegeneration such as plasma NfL often occurs only in the final months leading up to clinical onset, consistent with the rapidity of disease after onset[23,24].

PrP lowering delays onset of prion disease in animal models. The advent of transgenic and knockout mice 25 years ago provided the first genetic evidence that PrP dosage governs the tempo of prion disease[11,25] (Figure 2A). These models are relevant because prion disease naturally afflicts not only humans but also diverse mammals including deer, elk, sheep, goats, cattle, camels, and mink. It is exceptionally well modeled in mice, where intracerebral inoculation of prions into naive wild-type animals leads to fatal disease within 5-6 months, with all of the same pathological hallmarks seen in the human disease[26]. Heterozygous knockout mice survive prion infection more than twice as long as wild-type mice[11] (Figure 2A), and homozygous knockouts are incapable of propagating prions or developing prion disease[8]. Conversely, disease is accelerated in transgenic mice overexpressing PrP[25]. This finding has replicated in conditional genetic models, where lowering or abolishing PrP expression can respectively delay or prevent development of disease[27,28]. More recently, we showed that chronic administration of an ASO lowering PrP by ~50% can triple survival in prion-infected mice[14] (Figure 2B). This result was obtained as long as therapy is initiated at any time from prior to infection up through 78 days post-infection (dpi), when animals are presymptomatic but neuropathological sequelae including astrogliosis and increase in plasma NfL are emerging[14]. Survival increase in early-treated animals was comprised mostly of an extension of healthy life. In animals treated at the cusp of frank symptom onset, a single dose of ASO was still able to increase survival by 70% (Figure 2C)[14]. Survival is directly responsive to PrP dosage: a low dose of ASO corresponding to 21% knockdown of *Prnp* RNA in cortex led to a significant 15% increase in survival time, but higher doses achieving deeper target engagement increased survival by larger amounts[14]. The efficacy of PrP lowering against prion disease in mice is dose-dependent, with no apparent minimum threshold to effect (Figure 2D). Pharmacologic reduction of PrP in the brain is also dose-dependently reflected in the CSF in rats (Figure 2E), suggesting the utility of CSF PrP by ELISA as a pharmacodynamic biomarker[17].

All of the proof-of-concept ASO studies described above were conducted with experimental rigor and transparency: animal monitoring and euthanasia decisions were taken by blinded technicians, all cause survival was reported as the primary endpoint, and individual-level animal data were publicly released along with source code sufficient to allow third parties to reproduce all of the reported statistics and figures (see https://github.com/ericminikel/prp_lowering/).

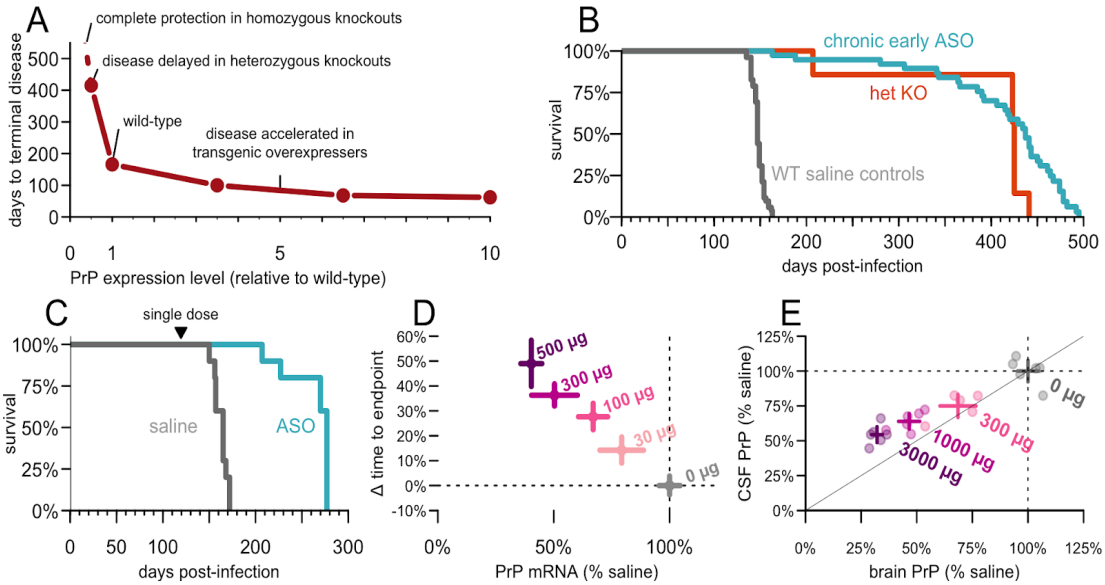

Figure 1. Published genetic and pharmacologic proofs-of-concept validating PrP lowering as a candidate therapy in prion disease. A) *Effect of PrP expression level on survival time after intracerebral prion inoculation in mice across wild-type, transgenic, and knockout lines.* ***B)*** *Impact of chronic early ASO treatment on a symptomatic endpoint mirrors heterozygous knockout in mice.* ***C)*** *A single ASO dose near the onset of symptoms delays terminal endpoint in mice.* ***D)*** *Continuous dose-response relationship between PrP and time to symptomatic endpoint in mice.* ***E)*** *CSF PrP mirrors brain PrP knockdown after ASO administration in rats. Panel A adapted from ref.* [25]*, B-D from ref.* [14]*, E from ref.* [17]*.*

Animal model and human genetic data suggest that PrP is non-essential and PrP lowering is well tolerated. Complete (homozygous) PrP knockout mice, cows, and goats are all healthy and viable and initially defied efforts to identify any knockout phenotype other than resistance to prion infection[29–32]. It was eventually discovered that homozygous (but not heterozygous) PrP KO mice exhibit a myelin maintenance deficiency in peripheral nerves (but not in the CNS) leading to a mild peripheral neuropathy[33]. This is apparently due to a lack of signaling of an N-terminal PrP peptide to Adgrg6, a receptor expressed on peripheral Schwann cells but not in the CNS[34]. While this myelin phenotype is histologically evident in aged animals, it is phenotypically subtle. Identifying a difference in grip strength, rotarod performance, or hot plate response is difficult, and some aged (60 week) PrP KO mice but not others exhibit a small but significant difference from wild-type animals, apparently as a function of mouse strain[33]. The histological phenotype has further replicated in PrP KO goats, where no outward phenotypic difference was observed[35]. Acute postnatal PrP lowering has also been observed to be safe in 2 conditional genetic mouse models[28,36] and with PrP-lowering ASOs[14]. We have identified older humans with naturally occurring heterozygous loss-of-function mutations in *PRNP* who appear to be healthy[2,37]. Moreover, such mutations occur in the population at exactly the frequency expected based on mutation rates; unlike the vast majority of genes in the human genome, there appears to be no purifying selection against loss-of-function variants in *PRNP*[37]. The fact that no homozygous *PRNP* knockout humans have yet been observed is expected based on mutation rates and the number of individuals sequenced, and thus does not constitute evidence that this

genotype is not tolerated[37]. Overall, animal and human genetic studies support that PrP lowering itself may be well tolerated in the context of this therapeutic approach.

CSF PrP can report on PrP lowering in brain and serve as a target engagement biomarker. In rats[17] (Figure 2E) and macaques[18], the lowering of brain PrP by ASOs is mirrored in CSF, suggesting that measurement of PrP in the CSF may serve as an adequate surrogate of brain target engagement for PrP-siRNA. In pre-symptomatic at-risk *PRNP* mutation carriers, CSF PrP is lower than in healthy controls without mutations[21,22], apparently due to altered metabolism and localization of the mutant protein[38]. Reduced CSF PrP in this population is not related to disease pathology because it is observed regardless of age and presence of pathological biomarkers, and these mutation carriers exhibit extremely stable concentrations of PrP in CSF over up to 6 years, with a within-individual coefficient of variation of just 9% over 6 years of follow-up[22]. Against such a stable long-term baseline in this population, an acute decrease in CSF PrP is expected to be readily detectable and report on drug target engagement.

2.2 Supporting Data

To supplement the evidence for PrP-lowering from genetic knockout and ASO experiments, we also sought to validate the therapeutic hypothesis with a divalent siRNA tool compound sharing the same chemical backbone as our clinical candidate. In vitro screening identified oligo 1682 as promising, with 11.7% residual *Prnp* mRNA. In vivo, at 30 days post-dose this compound at 10 nmol (348 µg) yielded 49.4% residual PrP protein in whole brain hemispheres measured by ELISA. This is similar to the potency of PrP-lowering ASO tool compounds we have tested previously. (Note that PrP-siRNA is more potent than this tool compound). We advanced the tool compound into studies in a standard disease model: wild-type mice infected intracerebrally with the RML strain of mouse prions. Divalent siRNA treatment was initiated at different timepoints in two cohorts: 75 days post-infection (dpi), corresponding to a pathological disease stage (evinced by elevated neurofilament and astrogliosis), and at 126 dpi, approximately at the onset of symptoms. 75 dpi animals received chronic treatment every 120 days (q120d) while 126 dpi animals received a single dose. At each dosing timepoint, animals received 348 µg of divalent siRNA split between bilateral doses into CSF via the cerebral ventricle. Control animals received a non-targeting control exNA compound, saline, or no injection.

In the 75 dpi early chronic study (Figure 3A),1682-exNA animals significantly outlived controls by 2.7-fold (median 442 vs. 165 dpi, P = 0.0002, log-rank test), with disease-attendant weight loss both delayed and slowed (Figure 3B). In the 126 dpi single late dose cohort, 1682-exNA animals lived 3 months longer than controls (median 270 vs. 164 dpi, P = 0.0002, log-rank test, Figure 3C), eventually developing typical prion disease signs and succumbing to weight loss endpoint (Figure 3D). This amounts to a 64% increase in *total* survival time, or a 3.8x increase in *remaining* survival time from the moment of treatment at 126 dpi. The impact of 50% reduction of PrP on disease progression appears generally similar whether achieved by di-siRNA, ASO, or heterozygous genetic knockout[14,15], consistent with our mechanistic understanding of prion disease as strictly dependent upon PrP expression.

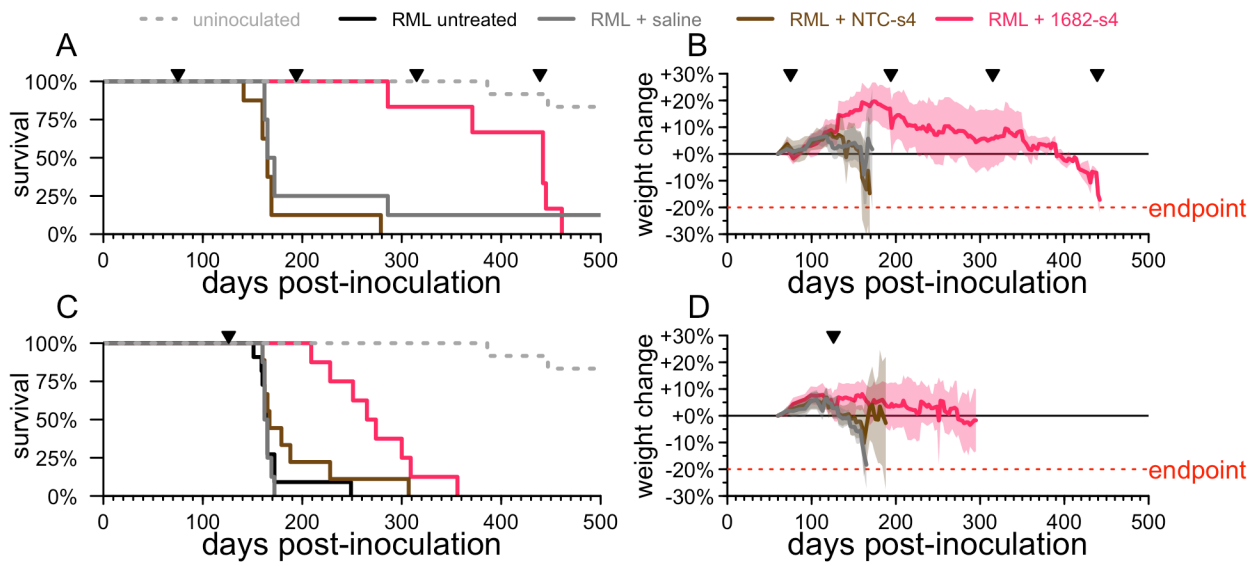

Figure 2. Efficacy of a divalent siRNA tool compound in mice with prion disease. A) *All-cause survival of RML prion-inoculated animals treated with 1682-exNA every 120 days beginning at 75 dpi, triangles indicate doses. N=8 per group.* ***B)*** *Weight change trajectories versus individual baseline in animals from (C), lines indicate means and shaded areas indicate 95%CIs.* ***C)*** *All-cause survival of RML prion-inoculated animals treated with a single dose of 1682-exNA at 126 dpi, triangle indicates dose. N=8-11 per group.* ***D)*** *Weight change trajectories versus individual baseline in animals from (E), lines indicate means and shaded areas indicate 95%CIs.*

Following extensive screening in cell culture and in transgenic mice expressing human PrP, we nominated PrP-siRNA as the lead candidate[15]. We performed a 3-point in vivo dose response experiment in human PrP mice with 5-fold dose increments (7, 35, and 174 µg) versus saline controls, N=8 per group, with harvest at 4 weeks. PrP protein in whole mouse brian hemispheres was dose-dependently reduced (Figure 3A). 6 mouse brain regions were dissected for regional qPCR of the *PRNP* transcript. Individual dose-response curves for each scaffold/tail combination in each brain region yielded median effective dose (ED_{50}) for PrP-siRNA ranging from 5 µg in thalamus to 18 µg in cerebellum (Figure 3B).

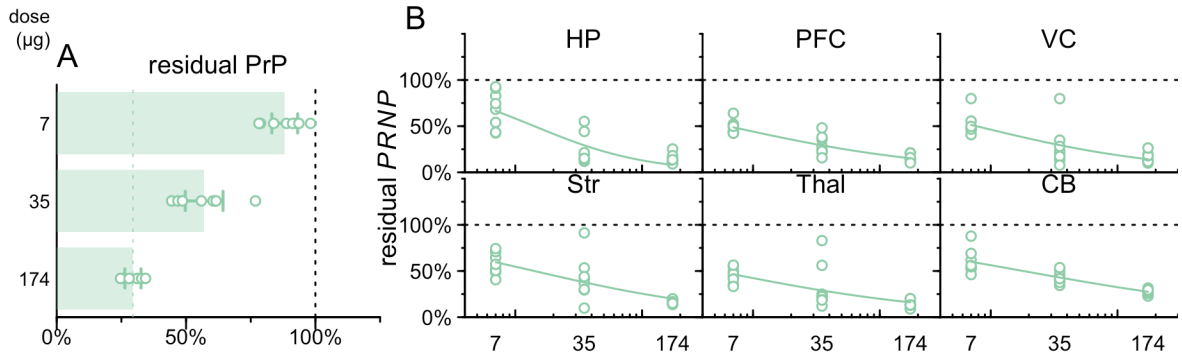

Figure 3. Dose-response of PrP-siRNA. A) *Whole hemisphere residual PrP by ELISA (x axis) by dose (y axis). Each point is one animal, rectangular bars are means, error bars are 95%*

confidence intervals. ***C)*** *Regional PRNP RNA by qPCR (y axis) versus dose (x axis). Each point is one animal. Curves are 4-parameter log-logistic dose-response curves.*

We tested the durability of effect of PrP-siRNA in 2 studies. After a 348 µg dose we observed 17% and 40% residual PrP in whole hemispheres at 34 and 120 days post-dose (Figure 4A). After a 174 µg dose we observed 29%, 51%, and 74% residual PrP at 29, 91, and 180 days post-dose (Figure 4B).

We also explored the effect of a loading dose administered 7 days after the initial dose, and of repeat dosing after 90 days (Figure 6C). When the 348 µg dose was given at day 0, 7, and 90, the residual PrP in whole hemisphere at day 120 was 15%. We also evaluated a 52 µg dose which may more realistically correspond to dose levels reachable clinically (see Discussion). Compared to a single dose at day 0, a loading dose regimen (day 0 and 7) for 52 µg provided improved target engagement at day 30 (32% vs. 49% residual PrP). An additional dose at day 90 yielded 41% residual at day 120.

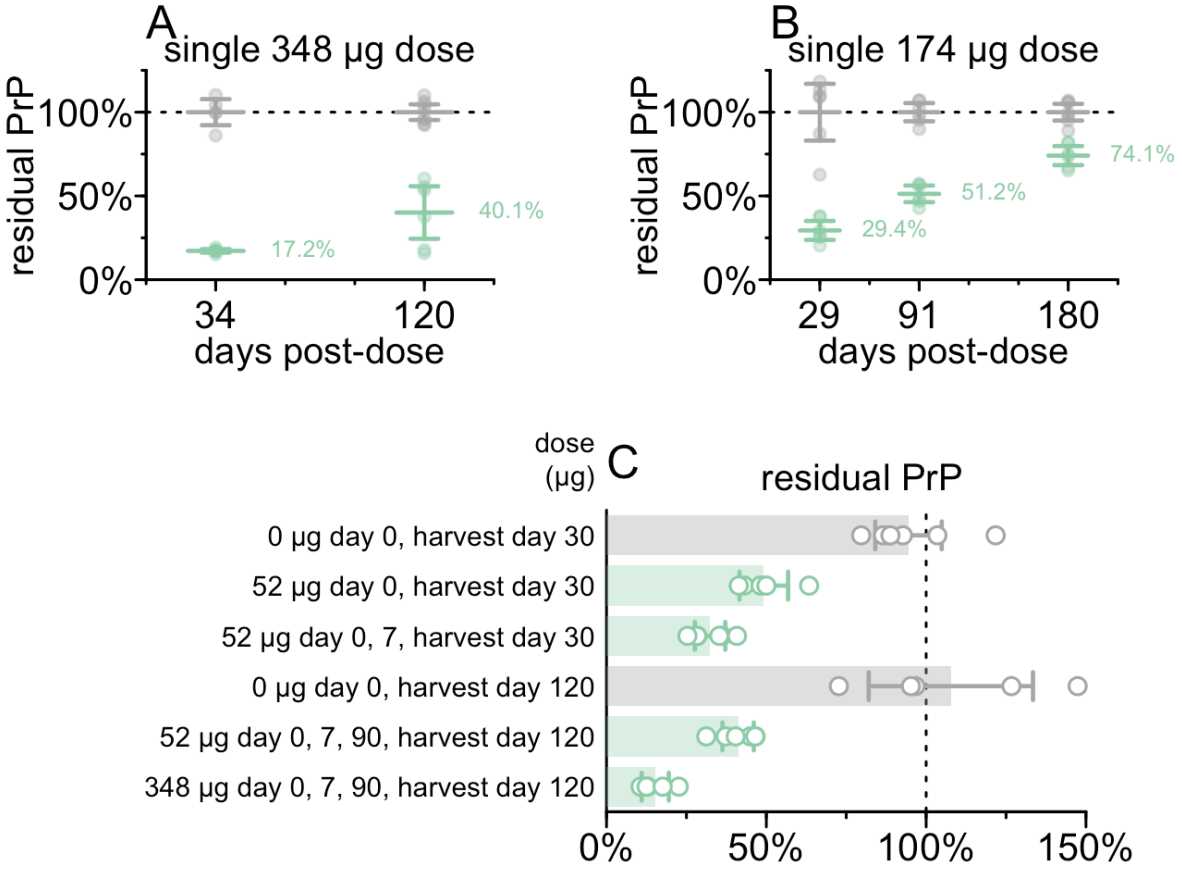

Figure 4. Characterization of lead compound PrP-siRNA. A) *Durability of effect of a single 348 µg dose in human PrP mice. Whole hemisphere PrP (y axis) versus months post-dose (x axis). Each point is one animal, horizontal line segments are means, error bars are 95% confidence intervals.* ***B)*** *Durability of effect of a single 174 µg dose in human PrP mice. Whole*

hemisphere PrP (y axis) versus months post-dose (x axis). Each point is one animal, horizontal line segments are means, error bars are 95% confidence intervals. ***C)*** *Impact of repeat dosing regimens on target engagement in human PrP mice. Whole hemisphere PrP (x axis) for indicated dose levels and regimens (y axis). Each point is one animal, rectangular bars are means, error bars are 95% confidence intervals.*

Note that the durability of PrP-siRNA, with activity out to 4 or 6 months, compares favorably with the median ~2 month survival time of prion disease patients in prior randomized trials[6,7].

2.3 Risk/Benefit Assessment

This drug has never been tested in humans before, and thus carries potential risks.

Nonclinical GLP toxicology studies were conducted by intrathecal administration of PrP-siRNA to rats and dogs, using the same GMP batch of material that will be used in the clinical trial. In both species, there were no in-life findings nor any pathology findings deemed adverse up to the maximum dose tested, which was 3 mg in rat and 200 mg in dog. In dogs given 200 mg, gliosis was observed at 24 hours post-dose in 4/8 animals, but was not associated with any necrosis or degeneration except focally at the injection site in 1 animal. In dogs given 60 mg, gliosis was observed in 1/8 animals at 24 hours post-dose. Gliosis was not observed in recovery animals at 28 days post-dose. Pathologists classified the findings as non-adverse. In rats, there were no pathological findings at any dose. Scaling based on CSF volume suggests that the maximum doses in these toxicology studies are approximately 8 times higher than the maximum dose planned for this clinical trial, and 32 times higher than the starting dose. Nevertheless, no siRNA of this chemical configuration has ever been administered to humans before, and chemical features of this investigational drug such as its tetraethylene glycol linker and its extended nucleic acid (exNA) internucleotide linkages could have toxicological properties in humans not anticipated from nonclinical toxicology studies.

PrP-siRNA is not sequence-matched to the PrP gene in rat nor in dog, thus the above studies cannot assess on-target toxicity due to PrP lowering. As detailed in Section 2.1, decades of evidence from PrP knockout animals (mice, cattle, goats) and recent evidence from heterozygous knockout humans all support the non-essentiality of PrP. Acute postnatal PrP lowering has also been observed safe in 2 conditional genetic models[28,36] and with PrP-lowering ASOs[14]. Nevertheless, no data from acute PrP lowering in humans are presently available, leaving a potential risk of side effects not anticipated from animal models.

Lumbar puncture and intrathecal drug administration are relatively safe procedures, however, minor and major complications can occur even when experienced and qualified proceduralists use proper technique. These potential complications include headache, back pain, bleeding, infection, radicular pain, cerebral herniation or the potential for epidermoid tumors of the thecal sac.

There are at present no human data to suggest a direct benefit to participants in this trial. Participants may benefit from the opportunity to contribute to research and generate generalizable knowledge about PrP-siRNA and about prion disease.

3 STUDY DESIGN

This is a first-in-human single ascending dose open label study of an intrathecal oligonucleotide drug in symptomatic patients diagnosed with prion disease.

Dose escalation and sentinel dosing. The dose levels will be 50, 100, and 200 mg, administered to 3, 3, and 9 patients. Each dose level will begin with 1 sentinel (see 5.1.5 for sentinel dosing plan).

Relationship between schedule of activities and study endpoints. As indicated in the Schedule of Activities (SOA), participants will complete a screening visit and then a baseline visit at which drug is administered. Hospital admission at the baseline visit will allow close monitoring of safety for at least 24 hours following drug administration. Adverse event reporting throughout the study will support the primary objective. Repeat lumbar punctures at 4-week intervals post-dose will allow monitoring of secondary endpoint pharmacodynamic biomarker CSF PrP, which is expected to decrease to its minimum level by 4 weeks post-dose as a result of drug treatment, then rebound gradually as drug washes out and to be within 20% of baseline by end of study at 24 weeks. Each participant's CSF PrP values will be compared to their own individual baseline, defined as the CSF PrP concentration at their baseline visit. For any participants who have previously participated in our observational cohort study of pre-symptomatic subjects[21,22], we will also compare to their historical samples gathered in that study. Blood and CSF collected at each visit will also permit measurement of the secondary endpoint of pharmacokinetics as well as exploratory biomarker endpoints. Clinical and imaging data collected at visits will also support exploratory objectives.

3.1 Scientific Rationale for Study Design

Rationale for intrathecal dosing. PrP-siRNA does not cross the blood-brain barrier. All proof-of-concept studies in animal models were conducted using ICV administration into CSF, and nonclinical GLP toxicology studies were performed using IT administration. Pharmacokinetic measurements in GLP toxicology animals indicate broad distribution of PrP-siRNA in the brain following intrathecal dosing.

Rationale for single dose. A single dose is supported by the single dose GLP toxicology studies conducted. This single dose study is a first step towards understanding the safety, tolerability, and pharmacology of the investigational drug in this population. (We are planning to complete repeat-dose toxicology studies to enable future trials, not part of this protocol).

Rationale for trial in pre-symptomatic patients. Our primary objective is to assess the safety and tolerability of PrP-siRNA in pre-symptomatic subjects at risk for genetic prion disease. Our secondary objective is to characterize pharmacokinetics and pharmacodynamic impact in this same population. Compared with a healthy volunteer study, it is important that PrP-siRNA is studied directly in at-risk individuals who have 1) mutant PrP with a different mRNA sequence and PrP amino acid sequence, 2) altered metabolism of mutant PrP resulting in reduced steady-state CSF concentrations of PrP at baseline[21,22]. Compared with a symptomatic patient

population (in which we are also planning a separate trial), it is important that we study at-risk individuals who 1) do not yet have clinical symptoms of prion disease, 2) do not yet have confounding changes in CSF PrP due to disease progression, and 3) do have PrP mutations (whereas a majority of symptomatic patients have sporadic disease with two wild-type *PRNP* alleles).

Rationale for open label design. In this small Phase I clinical trial, our power to characterize common adverse events and to detect rare adverse events is maximized if all participants are treated with investigational drug. The stable baseline concentration of CSF PrP in this population makes it feasible to assess target engagement by comparing each individual to their own pre-treatment baseline, without needing to include a placebo arm.

Rationale for trial duration. In preclinical proof-of-concept studies, PrP lowering by PrP-siRNA was maximal at 1 month post-dose and PrP had begun to rise back towards its baseline by 3-4 months. Even at a dose higher than those tested in this trial, pharmacodynamic activity was limited by 6 months post-dose. Thus, the 24-week trial duration is anticipated to span the pharmacodynamic period of PrP-siRNA and thus when drug-related adverse events are most likely to occur.

Rationale for inclusion and exclusion criteria. Because our objective is to evaluate PrP-siRNA in pre-symptomatic subjects at genetic risk for prion disease, we require that patients not have any symptoms of prion disease (a separate trial will target those individuals with symptoms); and we require the presence of a highly penetrant *PRNP* genetic variant. We have chosen not to include individuals with lower penetrance variants because they are unlikely to develop disease in their lifetime and therefore are not the target population for longer term clinical development of this investigational drug.

3.2 Justification for Dose

Our proposed starting dose for our first in human phase 1 clinical trial for PrP-siRNA in pre-symptomatic at-risk subjects is a single dose of 50 mg intrathecally administered. We have chosen a dose that allows reasonably rapid attainment of our phase 1 trial objectives: assessment of the investigational drug's safety, tolerability, or pharmacokinetic and pharmacodynamic profiles.

The route of administration is intrathecal. Oligonucleotide drugs do not cross the blood-brain barrier and are not available in the brain following systemic delivery. Delivery into CSF results in broad, albeit non-uniform, distribution throughout the CNS. This has been demonstrated clinically with ASOs and in nonclinical models for our divalent siRNA. Intrathecal administration of the clinical drug product 2349-exNA is consistent with the intrathecal administration performed in animal toxicology studies completed to support its safety in nonclinical test species.

The dosing regimen is single dose. A single dose is supported by our single dose toxicology studies in rats and dogs. Understanding the tolerability and pharmacology of a single dose is a first step towards eventually studying a repeat dosing regimen in this population. Going forward,

we do plan to conduct repeat dose toxicology studies, and indeed, proposed designs for these studies are being submitted to FDA as part of this IND package (see 2.4 Nonclinical Overview > Planned Nonclinical Studies).

We propose a starting dose of 50 mg, with escalation to 100 mg and 200 mg. In setting a starting dose, we considered the margins both by CSF volume calculation and by local injection concentration (dose formulation concentration) as suggested by communications from FDA[39]. In Table 1, we compare our proposed dose levels in humans to the NOAELs for rat and dog, using CSF volumes from Emami et al[39,40].

Table 1. Proposed dose levels compared to toxicology data.

species	CSF volume (mL)	dose volume (mL)	NOAEL (mg)	proposed doses (mg)	local concentration (mg/mL)	CSF concentration (mg/mL)	margin by local vs. rat	margin by local vs. dog	margin by CSF vs. rat	margin by CSF vs. dog
rat	0.25	0.03	3	–	100.0	12.0				
dog	16	3	200	–	66.7	12.5				
human	130	20	–	50	2.5	0.4	40.0	26.7	31.2	32.5
				100	5.0	0.8	20.0	13.3	15.6	16.3
				200	10.0	1.5	10.0	6.7	7.8	8.1

At the proposed starting dose of 50 mg, we have at least a 25-fold safety margin both by local concentration and by CSF concentration compared to the NOAELs from both species. At the proposed maximum dose of 200 mg, we have at least a 6.7-fold safety margin by all four calculation methods.

Additional considerations justifying the dose are as follows.

Our qPCR data from humanized mice treated with a dose-response series of our drug indicates ED_{50} values of 5 – 18 µg across 6 brain regions, which based on CSF volume scaling (130 mL human vs. 0.04 mL mouse, a factor of 3,250) would correspond to 59 mg in a human, thus, our starting dose level is in a range expected to have meaningful pharmacologic activity. We know from animal models of disease that 50% PrP lowering slows but does not arrest disease; thus we wish to achieve deeper than 50% lowering, and we therefore intend to escalate to the 200 mg dose, predicted to achieve deeper than 50% knockdown.

We do not intend to go above 200 mg due to Drug Product formulation constraints. We formulated our Drug Product at 10 mg/mL. Data from antisense oligonucleotides establish 20 mL as a valid dose volume for intrathecal oligonucleotides; data on volumes larger than this are lacking. Thus, with our current formulation it is not possible to dose more than 200 mg in one administration. Future development of PrP-siRNA, not included under this submission, could include repeat dosing, to be informed by repeat dose toxicology studies as well as by target engagement durability results from this initial first in human single dose clinical trial.

Finally, our proposed doses meet endotoxin requirements. Testing of our Drug Product yielded a result of 0.2 EU/mL endotoxin (better than the acceptance criterion, which was 0.5 EU/mL). For the worst case of a 50 kg patient receiving a 200 mg drug dose in a 20 mL volume, this is 20 mL * 0.2 EU/mL / 50 kg = 0.08 EU/kg endotoxin. FDA permits up to 0.2 EU endotoxin per kg of patient body weight intrathecally in a single dose. Thus, our maximum dose fits within established endotoxin limits.

3.3 Study completion and end of study definitions

A participant is considered to have completed the study if he or she has completed all visits shown in the Schedule of Activities (SOA).

The primary completion date is defined as completion of the last visit or procedure shown in the SOA for the last participant.

4 SELECTION AND ENROLLMENT OF SUBJECTS

4.1 Inclusion Criteria

1. Positive genetic test for a highly penetrant *PRNP* mutation. These are mutations where either de novo occurrence in prion disease cases or Mendelian segregation in families have been observed, indicating that a large majority of individuals harboring this mutation eventually develop prion disease. The high penetrance of these mutations is further supported by their high frequency in case series relative to their very low frequency in, or total absence from, general population cohorts[2]. See Goldman & Vallabh 2022 [ref. [19]] Table S1 for a review of the literature evidence supporting the penetrance of such mutations.
2. No symptoms of prion disease in the opinion of the Investigator.
3. Age ≥18 at the time of informed consent.
4. Meeting the criteria for prevention of pregnancy.
5. Written informed consent and be able and willing to comply with all study requirements (in the opinion of the Investigator), including travel to the Study Center, procedures, assessments and visits.

4.2 Exclusion Criteria

1. Symptoms of active prion disease or other concomitant non-prion brain disease.
2. Attempted suicide, suicidal ideation with a plan that required hospital admission and/or change in level of care within 12 months prior to Screening. For patients with (i) a suicide ideation score ≥ 4 on the C-SSRS within the last 12 months, or (ii) suicidal behaviors within the last 12 months (as measured by the answer “Yes” on any of the C-SSRS Suicidal Behavior Items), a risk assessment should be done by an appropriately qualified mental health professional (e.g., a Psychiatrist or licensed Clinical Psychologist) deemed by the Investigator to be at

significant risk of suicide, major depressive episode, psychosis, confusional state or violent behavior should be excluded.
3. Platelet count < 50,000/mm3 , INR>1.5 or any other clinically significant laboratory abnormality at Screening that, in the opinion of the Investigator, would render a patient unsuitable for inclusion.
4. Clinically significant vital sign or electrocardiogram (ECG) abnormality at Screening (including heart rate < 45 bpm; SBP < 90 mmHg; confirmed BP readings > 170/105 mmHg).
5. Active infection requiring systemic antiviral or antimicrobial therapy that will not be completed prior to Study Day 1.
6. Active infection with hepatitis C, hepatitis B or HIV diagnosed by initial serological testing and confirmed with RNA testing, or prior treatment for hepatitis C. Patients at Screening who test positive by serology, but negative by RNA may be deemed eligible by the Investigator in consultation with the IMM.
7. Malignancy within 5 years, except for basal or squamous cell carcinoma of the skin or carcinoma in situ of the cervix that has been successfully treated. Patients with a history of other malignancies that have been treated with curative intent and which have no recurrence within 5 years may also be deemed eligible by the Investigator in consultation with the IMM.
8. Participation in a different clinical trial of experimental (approved or investigational) therapy for prion disease within 4 months prior to Screening.
9. Unwillingness to comply with study procedures, including follow-up, as specified by this protocol, or unwillingness to cooperate fully with the Investigator.
10. Concomitant treatment with immune modulating therapy and/or diagnosis of a disorder that may require treatment with immune modulating therapy including, but not limited to, autoimmune disorders.
11. Previous treatment with an oligonucleotide (including siRNA) within 4 months of screening. (This does not apply to mRNA vaccines, including COVID-19 vaccines.)
12. History of bleeding diathesis or coagulopathy, or current or anticipated use of warfarin, apixaban or other anti-coagulant or anti-platelet medication that cannot be safely held for the appropriate time prior to lumbar puncture.
13. Any contraindication to repeated lumbar punctures, including bleeding disorders, brain or spinal diseases or deformities. Examples include but are not limited to mass lesions, obstructive hydrocephalus, ventriculoperitoneal shunt, Chiari malformations, significant spondyloarthropathy and spinal stenosis, significant connective tissue disorders.
14. Any contraindication or unwillingness to undergo MRI (e.g., metal implants, claustrophobia, agitation or motor symptoms that precludes scans).
15. Have any other condition, which, in the opinion of the Investigator would make the patient unsuitable for inclusion or could interfere with the patient participating in or completing the study.
16. Patient unwilling to comply with the contraceptive requirements during the study period.

    Females of child-bearing potential must be willing to use a highly effective method of contraception from 30 days before dose administration until 6 months after dose administration or until study completion.

    Birth control methods which are considered highly effective include bilateral tubal occlusion, placement of an intrauterine device or intrauterine hormone-release system,

surgical sterilization of male partner, established use of oral, implantable, injectable or transdermal hormonal methods of contraception.

A female of child-bearing potential includes individuals who have experienced menarche and who are not postmenopausal or permanently sterilized. A post-menopausal state is defined as the absence of menses for 12 months without an alternative medical cause.

Males with partners of child-bearing potential must be surgically sterilized or agree to use a condom throughout study participation.

17. Female patient is breastfeeding, pregnant or planning a pregnancy.

4.3 Lifestyle Considerations

There are no lifestyle restrictions.

For prohibited medications see section 5.4.2.

4.4 Participant Study Enrollment Procedures

4.4.1 Participant Recruitment and Retention

Postings will be placed on Prion Registry (prionregistry.org) and with participating patient advocacy organizations such as Prion Alliance and CJD Foundation.

Dr. Steven Arnold is a neurologist in the MGH Memory Disorders Unit and subjects may also be recruited from among his patients. Special care will be taken to ensure patients are aware that their participation is completely voluntary and that their decision to participate will not affect their care. As with other clinicians in the Memory Disorders Unit, if the investigator has identified a patient as a potential candidate for the study, the investigator will either present the patient with a recruitment letter or inform the patient of the presence of the study coordinator, who will be present in a private waiting area of the Wang Ambulatory Neurology clinic. If the patient approaches the study coordinator, the study coordinator will discuss recruitment with the subject and their family member at the end of a scheduled clinic visit. The research coordinator will ask the patient if they would like to be contacted to hear about potential clinical trials. The coordinator will collect the patient's contact information and add it to a shared database. After the patient has been deemed potentially eligible to participate in this study by a doctor. A study coordinator will contact the subject and explain study in further detail and if the subject is interested, potentially complete a telephone prescreening.

Finally, subjects who have undergone screening for other protocols in the Alzheimer's Clinical & Translational Research Unit (ACTRU) directed by Dr. Arnold may be contacted about participating in this trial. Only subjects who have indicated that they are open to future contact

about other study participation opportunities on telephone prescreens and consent forms will be contacted in this manner.

4.4.2 Screening Documentation

Data related to eligibility or reasons for ineligibility for participants who have consented to participate will be maintained in the in the Electronic Data Capture System.

4.4.3 Informed Consent

Written informed consent will be obtained from each study participant before any study-specific procedures or assessments are done and after the aims, methods, anticipated benefits, and potential hazards are explained. The consent form will be signed by the participant with the date of that signature indicated. The investigator will keep the original consent form(s) and a copy will be given to the participant. It will also be explained to the participant that they are free to refuse entry into the study and free to withdraw from the study at any time without prejudice to future treatment. Written and/or oral information about the study in a language understandable by the participant will be given to all participants.

Informed consent will be obtained by licensed staff who are trained on the study and listed on delegation logs as being authorized to obtain consent. The investigator, or other licensed staff, will be involved in explaining the study and will play a role in the consent process, answering any questions in conjunction with the coordinator and ensuring that the process is carried out correctly. All subjects will be offered the opportunity to discuss participation with a licensed physician Investigator and the subject's decision to accept or decline this opportunity will be documented in the research file.

4.4.4 Assent

All participants must be cognitively intact and provide their own informed consent. Assent is not applicable.

4.4.5 Randomization/Treatment Assignment

This is an open-label study. All participants will receive PrP-siRNA.

4.5 Participant Withdrawal Criteria

Subjects are free to withdraw from participation in the study at any time upon request. Participants who withdraw will not be replaced.

5 STUDY INTERVENTIONS/INVESTIGATIONAL AGENT OR DEVICE

5.1 Investigational Agent/Interventions, Administration, and Duration

5.1.1 Blinding

All patients will receive PrP-siRNA and the dose administered is unblinded. There is no blinding in this open label study.

5.1.2 Setting

Investigational drug administration will take place at the clinical study center. Participants will be admitted to the hospital for at least 24 hours of post-dose monitoring.

5.1.3 Dosing instructions

The injection site will be lumbar intrathecal, between L4 and L5 unless the study Investigator's judgment determines that a different lumbar location is necessary. Non-lumbar dosing will not be permitted. The individual performing the lumbar puncture (principal investigator or experienced nurse practitioner) will first swab the injection site with povidone/iodine and may provide local anesthesia by injecting subcutaneous lidocaine, or similar, as deemed appropriate by the interventionalist. Upon insertion of an atraumatic spinal needle, they will collect 20 mL of CSF. The syringe received from the Research Pharmacy containing the investigational drug (20 mL volume) will then be attached to the spinal needle, and the drug product will be ejected under gentle pressure over the course of approximately 2 minutes.

Predose CSF collection of 20mL prior to injection of 20mL of investigation product ensures that the net administration procedure is isovolumetric. The CSF collected will be processed and stored for analysis.

5.1.4 Dosing frequency

Each participant will receive only a single dose of drug.

5.1.5 Sentinel dose and dose escalation scheme

Three dose level cohorts will be enrolled sequentially, with each participant receiving a single dose of either 50, 100, or 200 mg. The number of participants per cohort will be 3, 3, and 9 patients respectively. Dose escalation in this trial will be tied to dose escalation in a parallel trial in symptomatic subjects with active prion disease. Each dose level will be studied in this trial only once 42 days have passed since the sentinel patient was dosed at that level in the symptomatic trial. In addition, the first participant in this trial at each dose level will serve as a separate sentinel for

this trial's population. If no stopping criteria (7.5) are reached within 14 days after dosing, PrP-siRNA may be administered to the remaining 2 participants in that dose group.

For each dose escalation, the decision to proceed to dosing the next cohort will be made by the Independent Medical Monitor (IMM), upon review of a minimum of 14 days of available post-dose safety data from the 2nd and 3rd participant of a dose cohort, a minimum of 28 days of available post-dose safety data from the 1st participant of a dose group, and a minimum of 42 days of available post-dose safety data from the 1st symptomatic participant in the same dose group in the parallel trial in symptomatic patients. If no stopping criteria have been reached, dose escalation is permitted.

5.2 Device Description and IDE Information

Not applicable; no device involved.

5.3 Handling of Investigational Agent/Interventions

The drug product is formulated at 10 mg/mL in 5.3 mL vials with guaranteed 5 mL extractable volume, thus, a 50 mg dose can be obtained from each vial. Vials will be shipped as needed from a GMP storage facility to the hospital's Research Pharmacy. The starting dose is 50 mg, and we plan to escalate to doses of 100 mg and 200 mg (see dose escalation scheme below). Regardless of the dose level in milligrams, the dose volume is 20 mL. Thus, for the 50 mg and 100 mg doses, vials of drug product will be diluted in Elliotts B (Lukare Medical) at the MGH Research Pharmacy. Regardless of dose level, a 20 mL syringe containing the indicated amount of drug will be provided by the Research Pharmacy to the investigator.

Because the investigational drug is administered intrathecally, sterility and maintenance of low endotoxin are essential. Syringes will be filled and dilutions performed in sterile aseptic conditions by experienced staff. The amount of Drug Product and diluent to be mixed will be according to the following table:

Dose level	Drug Product	Diluent	Total Volume
50 mg	5 mL	15 mL	20 mL
100 mg	10 mL	10 mL	20 mL
200 mg	20 mL	—	20 mL

The time from removal of Drug Product from -20°C cold storage to intrathecal injection shall not exceed 8 hours. During this time, the Drug Product and/or syringe containing investigational drug with or without diluent may be stored at room temperature.

In accordance with local regulatory requirements, the investigator, designated site staff, or head of the medical institution (where applicable) must document the amount of investigational product dispensed and/or administered to study participants, the amount received from the central pharmacy, and the amount destroyed upon completion of the study. An investigator is responsible for ensuring that product accountability records are maintained throughout the course of the study.

The inventory will include details of vials received and dispensed to participants, batch, and ID numbers. All unused vials must be kept until the site receives written instruction from the monitor to destroy the remaining investigational agent/device or to return it to the Central Pharmacy or manufacturer. An accounting must be made of any drug deliberately or accidentally destroyed. Discrepancies between the amount of drug product received and dispensed drug must be reconciled.

5.4 Concomitant Interventions

No concomitant interventions are required and none will be provided.

5.4.1 Required Concomitant Medications/Interventions

No concomitant medications are required and none will be provided.

5.4.2 Prohibited Medications/Interventions

PrP-siRNA is not an inhibitor, substrate, or potentiator of any known enzymes or transporters. A drug-drug interaction study revealed no evidence of any interactions. Anti-coagulant / anti-platelet drugs (other than low-dose aspirin) are prohibited because they would increase the risk of adverse events related to the lumbar puncture. If an individual in the study has a reason to begin treatment with one of these drugs, they will be asked to discontinue the follow-up lumbar punctures, but they will be asked to continue repeated study visits if possible to gather safety, blood biomarker and neurocognitive assessment data.

5.4.3 Precautionary Medications/Interventions

Not applicable.

5.5 Participant compliance

Not applicable; this is a one-time intrathecal drug administration performed by study staff.

6 CLINICAL AND LABORATORY EVALUATIONS/STUDY PROCEDURES

6.1 Schedule of Activities

The study's Schedule of Activities is shown below.

Activity	**Pre-Screen**	**Screening**	**Baseline / drug admin & monitor**	**Week 4**	**Week 8**	**Week 12**	**Week 24**
	Phone	Clinic	Clinical study center	Clinic	Clinic	Clinic	Clinic
		Day -7±7[1]	Days 0-1	Day 28±7	Day 56±7	Day 84±7	Day 168±3
Pre-Screen Questionnaire & T-MoCA	1						
Written Informed Consent		1					
Demographics		1					
Neurocognitive & Medical History		1					
Inclusion/Exclusion Review		1					
Vital Signs		1	1	1	1	1	1
Physical Examination, Neurological Exam		1	1	1	1	1	1
Adverse Event Review		1	1	1	1	1	1
Concomitant Medication Review		1		1	1	1	1
Columbia-Suicide Severity Rating Scale		1		1	1	1	1
12-lead ECG monitoring		1	1				
Clinical study center admission			1				
Blood Collection		1	1	1	1	1	1
DNA Collection		1					
CSF Collection			1	1	1	1	1

Clinical Safety Labs		1	1	1	1	1	1
Brain MRI		1			1		
MRC-PDRS		1	1	1	1	1	1
MoCA		1		1	1	1	1
Administer/Dispense Investigational Product			1				
Exit Questionnaire							

[1]*All screening activities will be completed prior to the baseline visit.*

Note that per section 4.1, presence of a highly penetrant *PRNP* genetic variant is an inclusion criterion for this study. The identification of this variant must be made clinically as standard of care, prior to pre-screen. Patients will not undergo screening for this study until they have already received this information clinically. Thus, turnaround time for genetic testing is not included in the interval between screening and baseline / drug administration.

The below sections define each of these activities and explain the justification for their inclusion in the study.

6.1.1 Pre-Screen Questionnaire and T-MoCA

Participants will be pre-screened over the phone or via videoconference by study staff to determine eligibility for screening and to ensure that the participant is safe to undergo an LP and other study assessments. These procedures include:

- Obtain prion disease diagnosis history, family history, and medical history
- Perform T-MoCA, an abbreviated telephone-based version of Montreal Cognitive Assessment, to determine cognitive impairment.
- Review and document concomitant medications and therapies
- Discuss willingness and desire to participate in a clinical trial and understanding of the difference between treatment and research

This activity is necessary for determining whether the prospective participant should be scheduled for a screening visit.

6.1.2 Written Informed Consent

Participants will be given adequate time to become familiar with the study and ask questions. An investigator will meet with the potential participant and their study partner to review the details of the study using the ICF as a guide. This discussion will include all the required elements of informed consent, including the purpose of the research, the procedures to be followed, the risks and discomforts, the potential benefit of contributing to generalizable knowledge, and alternative

procedures to study participation. Their questions will be answered to their satisfaction. The Participant will be provided with adequate time to reflect on the potential benefits and risks and possible discomforts of participation, and to make an informed decision. A trained study coordinator or investigator will then obtain written informed consent from each participant prior to the initiation of any study procedures. All potential participants will be given the opportunity to speak with the site Investigator per current IRB policies should a non-Physician investigator be involved in obtaining informed consent.

This activity is necessary for enrollment into the study.

6.1.3 Demographics

Sex, age, and self-reported race and ethnicity will be collected.

This activity is necessary for record keeping.

6.1.4 Neurocognitive & Medical History

A licensed Nurse Practitioner Investigator or Physician Investigator will take a detailed medical history, particularly regarding neurological and cognitive symptoms and the diagnosis of prion disease.

This activity is necessary for determining whether the participant meets inclusion/exclusion criteria.

6.1.5 Inclusion/Exclusion Review

A trained study coordinator or investigator will interview the participant and study partner about each item in the inclusion/exclusion list in order to assist in determining eligibility.

This activity is necessary for determining whether the participant meets inclusion/exclusion criteria.

6.1.6 Vital Signs

Vital signs will be obtained after the participant has been in a seated position for a minimum of five minutes. Vital signs, including systolic and diastolic blood pressure, pulse rate /minute, respiratory rate/minute, temperature, and weight will be assessed at the visit. Height will be measured and recorded at the first visit only.

This activity is necessary for determining whether the participant meets inclusion/exclusion criteria.

6.1.7 Physical Examination, Neurological Exam

Physical and neurological exams will be performed by the site Investigator or other licensed clinical co-investigator. The following systems will be examined; general appearance, head, eyes, ears, nose, throat, neck, chest, heart, abdomen, extremities, edema, peripheral vascular, skin and appendages, musculoskeletal, and back. Neurological examination will occur at all study visits and will consist of assessment of mental status, cranial nerves, motor and somatosensory functions, reflexes, cerebellar function and gait.

This activity is necessary for safety monitoring.

6.1.8 Adverse Event Review

A licensed Nurse Practitioner Investigator or Physician Investigator will interview the participant and their study partner about any adverse events experienced since the last visit, and review the status or resolution of any previously reported averse events. At the screening visit, adverse events that occur after signing the consent form will be recorded.

This activity is necessary for the study's primary objective of determining safety and tolerability.

6.1.9 Concomitant Medication Review

A licensed Nurse Practitioner Investigator or Physician Investigator will interview the participant and their study partner about the participant's concomitant medications.

This activity is necessary for inclusion/exclusion criteria.

6.1.10 Columbia-Suicide Severity Rating Scale

The US FDA recommends the use of a suicidality assessment instrument that maps to the Columbia Classification Algorithm for Suicide Assessment (C-CASA)[41]. The C-CASA was developed to assist in coding suicidality data accumulated during the conduct of clinical trials of antidepressant drugs. The Columbia Suicide Severity Rating Scale (C-SSRS) will be utilized in this trial. The C-SSRS involves a series of questions designed to detect possible suicidal thinking and behavior. The full version C-SSRS (lifetime, and within the last 6 months) will be administered at the screening visit. At subsequent visits we will only ask about the last 6 months.

This activity is necessary for inclusion/exclusion criteria and for the study's primary objective of determining safety and tolerability.

6.1.11 Phlebotomy

Blood will be collected by standard phlebotomy at each visit. Approximately 40 mL will be obtained and processed for plasma, serum, buffy coat and, if applicable, DNA.

Blood tests required for safety prior to enrollment in the study may be performed using CLIA waived point of care testing using the Sysmex XW-100 CBC analyzer, Abaxis Piccolo XPress chemistry analyzer, and the Roche CoaguChek XS Plus PT/INR analyzer.

This activity is necessary for confirming eligibility per the inclusion/exclusion criteria, for performing clinical safety labs (see below) that inform the study's primary objective of evaluating safety and tolerability, as well as for secondary and exploratory endpoints.

6.1.12 DNA collection

At the time of phlebotomy during the screening visit, a tube of blood will be obtained for DNA extraction.

This activity is necessary for confirming eligibility per inclusion/exclusion criteria, and genotype information may also be used in stratification analyses related to any of the study's endpoints.

6.1.13 CSF collection

The injection site will be lumbar intrathecal, between L4 and L5 unless the study Investigator's judgment determines that a different lumbar location is necessary. The individual performing the lumbar puncture (principal investigator or experienced nurse practitioner) will first swab the injection site with povidone/iodine and may provide local anesthesia by injecting subcutaneous lidocaine, or similar, as deemed appropriate by the interventionalist. Upon insertion of an atraumatic spinal needle, they will collect 20 mL of CSF. The CSF collected will be processed, stored, and analyzed for drug concentration (pharmacokinetics), PrP concentration (pharmacodynamic biomarker), and markers of prion disease, neurodegeneration, and neuroinflammation.

This activity is necessary for the study's pharmacokinetic and pharmacodynamic secondary endpoints and for its exploratory endpoints. In the case of the day of dosing, CSF collection is also necessary to ensure isovolumetric drug administration.

6.1.14 Clinical Safety Labs

The following safety labs will be conducted, with rationale as provided for each:

- PT/INR. Coagulation in the normal range is essential in order for a lumbar puncture to be safe.
- Complete blood count (CBC) with differential. Although never observed for divalent siRNA, some other oligonucleotide drugs (ASOs) have been associated with thrombocytopenia.

- Comprehensive metabolic panel. Oligonucleotide drugs are generally metabolized through liver and kidney. If liver enzymes become elevated or kidney function declines, this could be indicative of drug toxicity.
- CSF cell count and differential. Increase in CSF cell count could be indicative of immune reaction or blood/brain barrier compromise.
- CSF glucose. CSF glucose could be indicative of changes in brain metabolism.
- CSF total protein. CSF total protein increase could be indicative of toxicity for an intrathecal drug or progression of underlying disease.

This activity is necessary for determining eligibility for lumbar puncture and for the study's primary objective of determining safety and tolerability.

6.1.15 12-lead ECG monitoring

The heart's electrical activity will be recorded from 12 electrodes placed on the skin. Cardiac arrhythmias are common drug side effects, and the ECG will allow monitoring for these after drug administration.

This activity is necessary for the study's primary objective of determining safety and tolerability.

6.1.16 Clinical study center admission

Admission to the clinical study center for at least 24 hours will allow careful monitoring of adverse events in the period immediately following drug administration.

This activity is necessary for the study's primary objective of evaluating safety and tolerability.

6.1.17 Brain MRI

A multi-sequence MRI scan will be conducted including the Alzheimer's Disease Neuroimaging scan protocol. We will compare structural and functional measures and brain volume between the screening visit and the week 8 visit.

This activity is necessary for the study's primary of evaluating safety and tolerability as well as for the exploratory objective of evaluating disease progression.

6.1.18 MRC-PDRS

The MRC-PDRS, or MRC Scale for short, measures cognitive function and ability to carry out activities of daily living. Because prion disease is very rapidly progressive and severe, other scales suffer from floor effects when rating the progression of prion disease patients. The MRC Scale was developed specifically to provide dynamic range throughout the progression of prion disease up to death[42].

This activity is necessary for the study's exploratory objective of evaluating clinical onset or markers of progression towards onset.

6.1.19 MoCA

The Montreal Cognitive Assessment (MoCA) is a commonly utilized questionnaire in clinical trials and research settings to measure levels of cognitive impairment. The MoCA measures five areas of cognitive function: orientation, visuospatial, attention and calculation, recall, and language. The MoCA will take approximately 10 minutes to complete. The MoCA will be administered only at Screening and Steady State Visit 4. Version 7.3 will be administered at Screening and version 8.3 will be administered at Week 12.

This activity is necessary for the study's exploratory objective of evaluating disease progression.

6.1.20 Administer Investigational Drug

Details of investigational drug administration are provided in section 5.1.3 above.

This activity is necessary for all study primary, secondary, and exploratory objectives.

6.1.21 Exit Questionnaire

This qualitative interview will gather information on participant perceptions of disease onset (if any) and experience of clinical trial participation.

This activity is necessary for the study's exploratory objective of evaluating clinical onset or markers of progression towards onset.

6.2 Timing of Study Activities

6.2.1 Timing of Activities and Study Visits

All screening activities may occur concurrent with one another, and all will be completed prior to drug treatment.

Allowable time windows for scheduling follow-up visits are shown in the SOA. If participants are unable to complete a visit within the indicate time window, study staff will do their best to accommodate a visit as close to the window as possible.

6.2.2 Investigational drug discontinuation procedures

Because this is a single-dose study, there are no discontinuation criteria for individual participants. Whether dosing of additional participants continues in the study will be determined by the stopping criteria in section 7.5.

6.2.3 On Study/Off-Intervention Evaluations

Because this is a single-dose study, there is no such thing as discontinuation of study drug. Individuals who have received drug will be expected to complete follow-up visits until/unless they choose to voluntarily withdraw from the study. Participants who choose to discontinue lumbar punctures will be encouraged to complete other aspects of remaining study visits to enable safety monitoring.

6.2.4 Off-Study Requirements

None. All follow-up visits are already included in the Schedule of Activities.

6.2.5 Pregnancy

Female patients who are breastfeeding, pregnant, or planning a pregnancy are excluded from the study as per Exclusion Criteria (see above).

Females of child-bearing potential must be willing to use a highly effective method of contraception from 30 days before dose administration until 6 months after dose administration or until study completion.

Birth control methods which are considered highly effective include bilateral tubal occlusion, placement of an intrauterine device or intrauterine hormone-release system, surgical sterilization of male partner, established use of oral, implantable, injectable or transdermal hormonal methods of contraception.

A female of child-bearing potential includes individuals who have experienced menarche and who are not postmenopausal or permanently sterilized. A post-menopausal state is defined as the absence of menses for 12 months without an alternative medical cause.

Males with partners of child-bearing potential must be surgically sterilized or agree to use a condom throughout study participation.

6.3 SPECIAL INSTRUCTIONS

6.3.1 Protocol Deviations

A protocol deviation is any noncompliance with the current clinical trial protocol. The noncompliance may be on the part of the subject, the PI or Co-Is, or the study staff. As a result of deviations, corrective actions will be developed by the PI and implemented promptly. All deviations from the protocol must be addressed in the subject's documents. Protocol deviations will be sent to the IRB per their guidelines and entered in the Protocol Deviations Log for the study.

6.3.2 Documentation of *PRNP* mutation

Presence of a highly penetrant pathogenic variant in *PRNP* will be obtained clinically as standard of care prior to screening. Results will be recorded in the participant's medical record.

6.3.3 Medical History

A medical history will be taken at the screening visit by study staff.

6.3.4 Treatment History

No effective therapeutic for prion disease exists; thus, no prior treatment is scientifically relevant to this study. Nevertheless, any treatment history reported by the participant will be included in the medical history collected by study staff at screening.

6.3.5 Concomitant Medications/Treatments

All concomitant medications, dietary supplements and/or other treatments should be documented.

6.3.6 Protocol Amendments and Study Termination

All revisions and/or amendments to this protocol must be approved in writing by the Sponsor and the IRB. The Investigator will not make any changes to the conduct of the study or the protocol without first obtaining written approval from the Sponsor and the IRB, except where necessary to eliminate an apparent immediate hazard to a study participant. The IRB will determine whether reconsent of participants is required after a protocol amendment.

The Sponsor and reserves the right to discontinue the study at a clinical study site(s) for safety or administrative reasons at any time. Should the study be terminated and/or the clinical study site closed for any reason, all documentation and investigational agent pertaining to the study must be returned to the Sponsor or its representative.

6.3.7 Laboratory Evaluations

The investigator will review laboratory test results and determine whether any qualify as adverse events; if so, they will be classified, and actions taken if indicated, per the instructions in section 7.2.

6.3.8 Participant Adherence Assessments

Participant adherence to the protocol is documented by their visits.

6.4 Details of laboratory endpoints

6.4.1 Pharmacokinetics

The investigational drug will be quantified in CSF and plasma using a mass spectrometry-based assay developed and validated at Battelle.

6.4.2 Pharmacodynamic biomarker

CSF total prion protein (CSF PrP) will be measured using an ELISA-based method developed at the Broad Institute[17] and validated at Battelle.

CSF for this endpoint will be collected and processed according to the same protocol established in our natural history study of presymptomatic individuals at risk for prion disease, in which pre-analytical variability was well-controlled and we observed longitudinal stability of CSF PrP concentration in the absence of drug treatment[22]. Studies using PrP-lowering antisense oligonucleotides in rats and cynomolgus macaques have established that CSF PrP reflects pharmacologic lowering of brain PrP[17,18]. Thus, we expect that lowering of brain PrP by PrP-siRNA should be quantifiable by serial measurement of PrP in CSF. Because each individual has a different baseline level of CSF PrP, each individual will be compared to the mean of their own pre-drug baselines, see statistical plan for details.

6.4.3 CSF RT-QuIC

CSF RT-QuIC is a seeded amplification assay that monitors fibrillization of recombinant PrP in a wellplate to determine whether CSF contains prion "seeds." RT-QuIC analyses will be performed to support the study's exploratory objective of evaluating clinical onset or progression towards onset.

6.4.4 Biomarkers of neurodegeneration and neuroinflammation

CSF NfL, T-tau, beta-synuclein, and plasma NfL, brain-derived tau, and GFAP will be quantified using Quanterix Single Molecule Immunoassay (Simoa) and/or ProteinSimple Ella assays. These will support the study's exploratory objective of evaluating clinical onset or progression towards onset.

6.4.5 DNA analysis

PRNP genotype impacts the age of onset and rate of disease progression and biomarker change in prion disease[20,43], and other variants genome-wide are reported to impact disease as well[44], so DNA sequencing will be performed to support exploratory endpoints and to enable exploratory post-hoc analyses of the primary and secondary endpoints stratified by *PRNP* genotype.

7 MANAGEMENT OF ADVERSE EXPERIENCES

The adverse event (AE) definitions and reporting procedures provided in this protocol comply with all applicable United States Food and Drug Administration (FDA) regulations and International Conference on Harmonization (ICH) guidelines. The Site Investigator will carefully monitor each participant throughout the study for possible AEs. All AEs will be documented on CRFs designed specifically for this purpose. It is important to report all AEs, especially those that result in permanent discontinuation of the investigational product being studied, whether serious or non-serious.

FDA and IRB requirements for reporting AEs will be followed. Participants will be monitored for AEs from the time they provide informed consent until 24 weeks following the one-time administration of study drug. At that point, all ongoing AEs will be followed to resolution, but no new AEs will be recorded. AEs that occur between consent and administration of study drug will be reported only if related to study procedures. At each visit (including telephone interviews), the participant will be asked "Have you had any problems or symptoms since your last visit?" in order to determine the occurrence of adverse events. If the participant reports an adverse event, the site Investigator will classify the AE per the definitions in section 7.1.2.

7.1 Definitions of Adverse Events, Suspected Adverse Drug Reactions & Serious Adverse Events

An adverse event (AE) is any unfavorable and unintended sign (including a clinically significant abnormal laboratory finding, for example), symptom, or disease temporally associated with a study, use of a drug product or device, whether or not considered related to the drug product or device.

Examples of adverse events include: new conditions, worsening of pre-existing conditions, clinically significant abnormal physical examination signs (i.e. skin rash, peripheral edema, etc), or clinically significant abnormal test results (i.e. lab values or vital signs), with the exception of non-safety related measurements, which are not being recorded as adverse events in this trial (they are being collected, but analyzed separately). Stable chronic conditions (i.e., diabetes, arthritis) that are present prior to the start of the study and do not worsen during the trial are NOT considered adverse events. Chronic conditions that occur more frequently (for intermittent conditions) or with greater severity would be considered as worsened and therefore would be recorded as adverse events.

Adverse events are generally detected in two ways:

Clinical → symptoms reported by the participant or signs detected on examination.

Ancillary Tests → abnormalities of vital signs, laboratory tests, and other diagnostic procedures (other than the non-safety related measurements: the results of which are not being captured as AEs).

7.2 Classification of adverse events

The different grades of adverse event are defined as follows:
Adverse events will be graded as:

- Mild
- Moderate
- Severe non-serious
- Serious not life-threatening / non-fatal
- Serious life-threatening / fatal.

The causality of adverse events may be defined as drug-related or not drug-related as follows:

Relatedness	Description
Not suspected drug related	Events consistent with natural history of disease and not following a temporal sequence from drug administration.
Suspected drug related	Events inconsistent with natural history of disease and/or following a temporal sequence from drug administration.

7.3 Assessment and Recording of Adverse Events

This study will utilize the CTCAE version 5.0 coding system for adverse event recording. Adverse events reported using CTCAE will be coded into MedDRA terms in the site Investigator or their study staff. Fields to be reported include:

- Type of event
- Date of onset and resolution (duration)
- Severity (mild, moderate, severe)
- Seriousness (does the event meet the above definition for an SAE?)
- Causality
- Action taken regarding investigational product
- Outcome

7.4 Reporting of adverse events

The site Investigator and research team are responsible for identifying adverse events, recording them, and complying with FDA and IRB reporting requirements for all safety reports. Copies of each report and documentation of IRB notification and receipt will be kept in the investigator's study file.

All clinical adverse events are recorded on the Adverse Event (AE) CRF and kept in the participant's study binder and must be entered into the online Adverse Event Reporting System **within 5 working days** of the site learning of a new AE or receiving an update on an existing AE.

Serious Adverse Events (SAEs) must be reported to the IMM **within 24 hours** of the site learning of the SAE. The IMM will review the SAE report and may request further information if necessary. The IMM may determine that the SAE requires expedited reporting to the FDA. If warranted, the IMM will prepare a Medwatch safety report for submission to the FDA.

7.5 Stopping criteria

The consequences of AEs will be determined according to the following contingency table:

AE Grade	Impact on Study Cohort
Mild	No effect.
Moderate	No effect.
Severe, non-serious	No effect.
Serious not life-threatening / non-fatal	If the AE is deemed related to PrP-siRNA in at least 1 subject: • dosing in that cohort will be suspended • dosing at a lower dose level cohort may continue • approval by the Sponsor, IRB and regulatory authority is required to resume dosing at the same or higher dose level. • if it is determined that dosing cannot be resumed at the same or higher dose level, additional patients can be enrolled at a lower dose level, up to the maximum of 15 patients total in this study.
Serious life-threatening / fatal	If the AE is deemed related to PrP-siRNA in at least 1 subject: • all dosing in the study will be stopped • approval by the Sponsor, IRB and regulatory authority is required to resume dosing at any dose level

As shown above, if a serious adverse event is suspected to be drug related, dosing must be halted. The Investigator must report the SAE to the IMM within 24 hours, and the IMM must report to the IRB and FDA within 7 days.

During periods when dosing is halted, or in the event that it is determined that dosing may not resume, participants who have already received their single dose of drug should continue to make follow-up visits for adverse event monitoring as well as to fulfill secondary and exploratory study objectives.

8 STATISTICAL CONSIDERATIONS

Statistical analysis will be carried out using the R programming language by the Sponsor and their staff. The statistical plan is detailed in the following sections.

8.1 General Design Issues

See section 3.1 for the overall rationale for this clinical trial design.

Each individual's CSF PrP concentration at each lumbar puncture will be compared that at baseline immediately prior to drug administration. Our longitudinal study of healthy controls and healthy asymptomatic individuals at risk for prion disease has established that individuals have different baseline concentrations of CSF PrP, driven primarily by *PRNP* genotype, but that any individual's CSF PrP is longitudinally stable.

Statistically, the approach is as follows. Each individual's baseline CSF PrP concentration will be defined as that obtained at the baseline / drug administration visit (day 0). Normalized CSF PrP at each subsequent timepoint will be expressed as a percentage of the individual's own baseline level. For the 4 and 8 week timepoints post-drug administration, a 2-sided paired T test will be used to compare the normalized CSF PrP concentrations in drug-treated individuals before and after treatment. This test will be used separately for the 50 mg, 100 mg, and 200 mg cohorts.

8.2 Sample Size Justification

8.2.1 Proposed sample size

We propose to include 15 patients, escalating dose as follows:

- 50 mg: 3 patients
- 100 mg: 3 patients
- 200 mg: 9 patients

We will be initiating repeat-dose toxicology studies sponsored by the URGenT program in parallel with the clinical study. If those studies are supportive, we aspire to initiate a subsequent repeat-dose study as a separate clinical trial.

8.2.2 Preliminary power analysis

In our longitudinal natural history study of healthy controls and pre-symptomatic at-risk individuals, we have monitored long-term (up to 6 years) within-subject test-retest reliability of CSF PrP and observed a coefficient of variation (CV) of 9.0%[22]. Our dose escalation scheme described above will yield a total of 9 patients treated at the highest tolerated dose. A 2-sided paired T test comparing pre- vs. post-dose CSF PrP levels will have 80% power to detect a 10% decrease in CSF PrP at the P=0.05 significance threshold.

8.3 Study Endpoints

8.3.1 Primary Endpoints

- Adverse events, classified per section 7.2.
- Complete blood count with differential.
- Complete metabolic panel.
- CSF glucose.
- CSF total protein.
- CSF cell count with differential.
- Vital signs including blood pressure, pulse, respiratory rate, temperature, and weight
- Physical exam parameters including general appearance, head, eyes, ears, nose, throat, neck, chest, heart, abdomen, extremities, edema, peripheral vascular, skin and appendages, musculoskeletal, and back.
- Neurological exam parameters including mental status, cranial nerves, motor and somatosensory functions, reflexes, cerebellar function, and gait.
- Suicidal assessment measured by C-CASA.

For details and justification of each of the safety labs, see section 6.1.14.

8.3.2 Secondary Endpoints

- CSF PrP (pharmacodynamic / target engagement biomarker).
- Mass spectrometry quantification of the drug in CSF and plasma (pharmacokinetics).

8.3.3 Exploratory Endpoints

- Survival
- MRC-PDRS
- CSF RT-QuIC
- CSF NfL
- CSF T-tau
- CSF beta-synuclein
- Plasma brain-derived (BD)-tau
- Plasma NfL
- Plasma GFAP

8.4 Data Analyses

8.4.1 Analysis Populations

The main analysis will include all participants. Post-hoc analyses may stratify by *PRNP* genotype.

8.4.2 Primary Analysis

Adverse events and serious adverse events will be tallied per participant and presented in tabular form.

8.4.3 Secondary Analysis

Each individual's baseline CSF PrP concentration will be defined that taken at the baseline / drug administration visit (day 0). Normalized CSF PrP at each subsequent timepoint will be expressed as a percentage of the individual's own baseline level. Statistical tests are described in section 8.

8.4.4 Exploratory Analysis

Exploratory data points will be plotted both in absolute terms (for instance, ng/mL or pg/mL) for neurodegeneration biomarkers) and relative to each individual's baseline.

8.4.5 Missing Data

Missing data will be omitted.

9 DATA COLLECTION and SITE MONITORING

9.1 Data Management

Site personnel will collect, transcribe, correct, and transmit the data onto source documents, CRFs, and other forms used to report, track and record clinical research data.

9.1.1 Registration

Registration of participants on this protocol will employ an interactive data system in which the clinical study site will attest to the participant's eligibility as per protocol criteria and obtain appropriate informed consent. Each participant will be assigned a de-identified 4-digit Participant Identification Number (PID).

9.1.2 Data Entry

Data will be digitized by study staff at the clinical site and stored in RedCap.

9.2 Quality Assurance

By signing this protocol, the Sponsor and Investigator agree to be responsible for implementing and maintaining quality control and quality assurance systems with written standard operating

procedures (SOPs) to ensure that studies are conducted and data are generated, documented, and reported in compliance with the protocol, accepted standards of GCP, and all applicable federal, state, and local laws, rules and regulations relating to the conduct of the clinical study.

9.2.1 Laboratory Data Flow

When a blood or urine sample has been obtained, the clinical study site study coordinator will send the sample (PID, site ID, and protocol ID numbers will be used) to the MGH central laboratory. Results will be sent via a secure system to MGH central laboratory with no individual-identifying information on the report. The laboratory will electronically communicate the test results to the respective clinical study sites in a secure manner.

9.3 Data and Safety Monitoring

All aspects of the study will be monitored by qualified individuals designated by the sponsor. Regulatory and data monitoring will be conducted according to Good Clinical Practice and applicable government regulations. The investigator agrees to allow monitors access to the Regulatory files, clinical supplies, dispensing and storage areas, and to the clinical files of the study participants, including electronic medical records, and, if requested, agrees to assist the monitors.

Safety monitoring will include careful assessment and appropriate reporting of adverse events by the study sites. Each Clinical Study Site Investigator and research team (co-Investigators, research nurse, clinical trial coordinator) are responsible for identifying and reporting AEs and determining the relationship of the event to the investigational agent/study procedures.

The Protocol PI will appoint an Independent Medical Monitor (IMM) to oversee the safety of the trial. The site Investigator will provide aggregate reports of all adverse events blinded by treatment group, detailed by severity, attribution (expected or unexpected), and relationship to the investigational agent/study procedures to the IMM on a quarterly basis. Serious adverse events not suspected to be drug-related will be reported to the IMM within 5 days. Serious adverse events suspected to be drug-related will be reported to the IMM within 24 hours, and dosing will be halted while the IMM reports to IRB and FDA and evaluates whether dosing should resume. The IMM is responsible for complying with IRB and FDA reporting requirements. The IMM will also evaluate whether the protocol or informed consent document requires revision based on the reports, and if so, the study staff will be responsible for making these revisions.
All adverse events in the reports will be coded using the MedDRA classification system.

10 HUMAN PARTICIPANTS

Documented approval from the IRB will be obtained for all participating sites prior to clinical trial start, according to ICH GCP, local laws, regulations and organization. When necessary, an extension, amendment or renewal of the IRB approval must be obtained.

Evidence of training in responsible conduct of research shall be on file for each study site Investigator and any co-investigators.

10.1 Institutional Review Board (IRB) Review and Informed Consent

This protocol and the informed consent document and any subsequent modifications will be reviewed and approved by the IRB responsible for oversight of the study. A signed consent form, approved by the IRB, will be obtained from the participant. For participants who cannot provide consent for themselves, such as those below the legal age, a parent, legal guardian, or person with power of attorney, must sign the consent form; additionally, the participant's assent, if applicable, must also be obtained if he or she is able to understand the nature, significance, and risks associated with the study. The consent form will describe the purpose of the study, the procedures to be followed, and the risks and benefits of participation. A copy of the consent form will be given to the participant, parent, or legal guardian, and this fact will be documented in the participant's record.

10.2 Participant Confidentiality

All laboratory specimens, evaluation forms, reports, video recordings, and other records that leave the clinical study site will be identified only by the study specific Participant Identification Number (PID) to maintain participant confidentiality. All records will be kept in a secure location. All computer entry and networking programs will be done using study specific PIDs only. Clinical information will not be released without written permission of the participant, except as necessary for monitoring by IRB, the FDA, the Sponsor, or the Sponsor's designee.

10.2.1 Certificate of Confidentiality

To further protect the privacy of study participants, a Certificate of Confidentiality will be issued by the National Institutes of Health (NIH). This certificate protects identifiable research information from forced disclosure. It allows the investigator and others who have access to research records to refuse to disclose identifying information on research participation in any civil, criminal, administrative, legislative, or other proceeding, whether at the federal, state, or local level. By protecting researchers and institutions from being compelled to disclose information that would identify research participants, Certificates of Confidentiality help achieve the research objectives and promote participation in studies by helping assure confidentiality and privacy to participants.

10.3 Study Modification/Discontinuation

The study may be modified or discontinued at any time by IRB, the Sponsor, FDA, or other government agencies as part of their duties to ensure that research participants are protected. If the study is terminated or suspended, the PI will promptly inform study participants, the IRB, and sponsor and provide the reason(s) for the termination or temporary suspension.

11 FUTURE USE OF STORED SPECIMENS AND DATA

CSF and blood samples from this study will be labeled with de-identified IDs and will be shipped to Eric Minikel's laboratory at the Broad Institute for analysis, storage, and future research purposes including studies conducted both on-site at the Broad Institute and studies conducted with outside collaborators. Results of any genetic testing performed on these samples will not be disclosed to participants.

12 STUDY RECORDS RETENTION

Records will be kept for 3 years from the primary completion date.

13 PUBLICATION OF RESEARCH FINDINGS

Publication of the results of this trial will be the responsibility of the Sponsor. Any presentation, abstract, or manuscript will be made available for review by the site Investigators prior to submission.

14 CONFLICT OF INTEREST POLICY

Any actual conflict of interest of persons who have a role in the design, conduct, analysis, publication, or any aspect of this trial will be disclosed and managed in accordance with their institution's policies. Furthermore, persons who have a perceived conflict of interest will be required to have such conflicts managed in a way that is in compliance with their institutional policy for management of conflicts of interest (COI).

15 REFERENCES

1. Prusiner SB. Prions. Proc Natl Acad Sci U S A. 1998 Nov 10;95(23):13363–13383. PMCID: PMC33918

2. Minikel EV, Vallabh SM, Lek M, Estrada K, Samocha KE, Sathirapongsasuti JF, McLean CY, Tung JY, Yu LPC, Gambetti P, Blevins J, Zhang S, Cohen Y, Chen W, Yamada M, Hamaguchi T, Sanjo N, Mizusawa H, Nakamura Y, Kitamoto T, Collins SJ, Boyd A, Will RG, Knight R, Ponto C, Zerr I, Kraus TFJ, Eigenbrod S, Giese A, Calero M, de Pedro-Cuesta J, Haïk S, Laplanche JL, Bouaziz-Amar E, Brandel JP, Capellari S, Parchi P, Poleggi A, Ladogana A, O'Donnell-Luria AH, Karczewski KJ, Marshall JL, Boehnke M, Laakso M, Mohlke KL, Kähler A, Chambert K, McCarroll S, Sullivan PF, Hultman CM, Purcell SM, Sklar P, van der Lee SJ, Rozemuller A, Jansen C, Hofman A, Kraaij R, van Rooij JGJ, Ikram MA, Uitterlinden AG, van Duijn CM, Exome Aggregation Consortium (ExAC), Daly MJ, MacArthur DG. Quantifying prion disease penetrance using large population control cohorts. Sci Transl Med. 2016 Jan 20;8(322):322ra9. PMCID: PMC4774245

3. Will RG, Ironside JW. Sporadic and Infectious Human Prion Diseases. Cold Spring Harb Perspect Med. 2017 Jan 3;7(1). PMID: 27793965

4. Pocchiari M, Puopolo M, Croes EA, Budka H, Gelpi E, Collins S, Lewis V, Sutcliffe T, Guilivi A, Delasnerie-Laupretre N, Brandel JP, Alperovitch A, Zerr I, Poser S, Kretzschmar HA, Ladogana A, Rietvald I, Mitrova E, Martinez-Martin P, de Pedro-Cuesta J, Glatzel M, Aguzzi A, Cooper S, Mackenzie J, van Duijn CM, Will RG. Predictors of survival in sporadic Creutzfeldt-Jakob disease and other human transmissible spongiform encephalopathies. Brain J Neurol. 2004 Oct;127(Pt 10):2348–2359. PMID: 15361416

5. Appleby BS, Rincon-Beardsley TD, Appleby KK, Crain BJ, Wallin MT. Initial diagnoses of patients ultimately diagnosed with prion disease. J Alzheimers Dis JAD. 2014;42(3):833–839. PMID: 24934543

6. Geschwind MD, Kuo AL, Wong KS, Haman A, Devereux G, Raudabaugh BJ, Johnson DY, Torres-Chae CC, Finley R, Garcia P, Thai JN, Cheng HQ, Neuhaus JM, Forner SA, Duncan JL, Possin KL, Dearmond SJ, Prusiner SB, Miller BL. Quinacrine treatment trial for sporadic Creutzfeldt-Jakob disease. Neurology. 2013 Dec 3;81(23):2015–2023. PMCID: PMC4211922

7. Haïk S, Marcon G, Mallet A, Tettamanti M, Welaratne A, Giaccone G, Azimi S, Pietrini V, Fabreguettes JR, Imperiale D, Cesaro P, Buffa C, Aucan C, Lucca U, Peckeu L, Suardi S, Tranchant C, Zerr I, Houillier C, Redaelli V, Vespignani H, Campanella A, Sellal F, Krasnianski A, Seilhean D, Heinemann U, Sedel F, Canovi M, Gobbi M, Di Fede G, Laplanche JL, Pocchiari M, Salmona M, Forloni G, Brandel JP, Tagliavini F. Doxycycline in Creutzfeldt-Jakob disease: a phase 2, randomised, double-blind, placebo-controlled trial. Lancet Neurol. 2014 Feb;13(2):150–158. PMID: 24411709

8. Büeler H, Aguzzi A, Sailer A, Greiner RA, Autenried P, Aguet M, Weissmann C. Mice devoid of PrP are resistant to scrapie. Cell. 1993 Jul 2;73(7):1339–1347. PMID: 8100741

9. Manson JC, Clarke AR, Hooper ML, Aitchison L, McConnell I, Hope J. 129/Ola mice carrying a null mutation in PrP that abolishes mRNA production are developmentally normal. Mol Neurobiol. 1994 Jun;8(2–3):121–127. PMID: 7999308

10. Prusiner SB, Groth D, Serban A, Koehler R, Foster D, Torchia M, Burton D, Yang SL, DeArmond SJ. Ablation of the prion protein (PrP) gene in mice prevents scrapie and facilitates production of anti-PrP antibodies. Proc Natl Acad Sci U S A. 1993 Nov 15;90(22):10608–10612. PMCID: PMC47826

11. Büeler H, Raeber A, Sailer A, Fischer M, Aguzzi A, Weissmann C. High prion and PrPSc levels but delayed onset of disease in scrapie-inoculated mice heterozygous for a disrupted PrP gene. Mol Med Camb Mass. 1994 Nov;1(1):19–30. PMCID: PMC2229922

12. Salvesen Ø, Espenes A, Reiten MR, Vuong TT, Malachin G, Tran L, Andréoletti O, Olsaker I, Benestad SL, Tranulis MA, Ersdal C. Goats naturally devoid of PrPC are resistant to scrapie. Vet Res. 2020 Jan 10;51(1):1. PMCID: PMC6954626

13. Raymond GJ, Zhao HT, Race B, Raymond LD, Williams K, Swayze EE, Graffam S, Le J, Caron T, Stathopoulos J, O'Keefe R, Lubke LL, Reidenbach AG, Kraus A, Schreiber SL,

Mazur C, Cabin DE, Carroll JB, Minikel EV, Kordasiewicz H, Caughey B, Vallabh SM. Antisense oligonucleotides extend survival of prion-infected mice. JCI Insight. 2019 30;5. PMID: 31361599

14. Minikel EV, Zhao HT, Le J, O'Moore J, Pitstick R, Graffam S, Carlson GA, Kavanaugh MP, Kriz J, Kim JB, Ma J, Wille H, Aiken J, McKenzie D, Doh-Ura K, Beck M, O'Keefe R, Stathopoulos J, Caron T, Schreiber SL, Carroll JB, Kordasiewicz HB, Cabin DE, Vallabh SM. Prion protein lowering is a disease-modifying therapy across prion disease stages, strains and endpoints. Nucleic Acids Res. 2020 Aug 10; PMID: 32776089

15. Gentile JE, Corridon TL, Serack FE, Echeverria D, Kennedy ZE, Gallant-Behm CL, Hassler MR, Kinberger G, Kamath NG, Lian Y, Gross KY, Miller R, DeSouza-Lenz K, Howard M, Guzman K, Chan N, Curtis DT, Fettes K, Lemaitre M, Cappon G, Jackson AL, Yamada K, Alterman JF, Coffey AA, Minikel EV, Khvorova A, Vallabh SM. Divalent siRNA for prion disease. bioRxiv. 2024 Dec 5;2024.12.05.627039.

16. An M, Davis JR, Levy JM, Serack FE, Harvey JW, Brauer PP, Pirtle CP, Berríos KN, Newby GA, Yeh WH, Kamath N, Mortberg M, Lian Y, Howard M, DeSouza-Lenz K, Guzman K, Thai A, Graffam S, Coffey AA, Frei J, Pierce SE, Safar JG, Deverman BE, Minikel EV, Vallabh SM, Liu DR. In vivo base editing extends lifespan of a humanized mouse model of prion disease. Nat Med. 2025 Jan 14; PMID: 39810005

17. Mortberg MA, Zhao HT, Reidenbach AG, Gentile JE, Kuhn E, O'Moore J, Dooley PM, Connors TR, Mazur C, Allen SW, Trombetta BA, McManus A, Moore MR, Liu J, Cabin DE, Kordasiewicz HB, Mathews J, Arnold SE, Vallabh SM, Minikel EV. Regional variability and genotypic and pharmacodynamic effects on PrP concentration in the CNS. JCI Insight. 2022 Mar 22;7(6):e156532. PMCID: PMC8986079

18. Mortberg MA, Gentile JE, Nadaf NM, Vanderburg C, Simmons S, Dubinsky D, Slamin A, Maldonado S, Petersen CL, Jones N, Kordasiewicz HB, Zhao HT, Vallabh SM, Minikel EV. A single-cell map of antisense oligonucleotide activity in the brain. Nucleic Acids Res. 2023 May 16;gkad371. PMID: 37188501

19. Goldman JS, Vallabh SM. Genetic counseling for prion disease: Updates and best practices. Genet Med Off J Am Coll Med Genet. 2022 Jul 12;S1098-3600(22)00812–7. PMID: 35819418

20. Minikel EV, Vallabh SM, Orseth MC, Brandel JP, Haïk S, Laplanche JL, Zerr I, Parchi P, Capellari S, Safar J, Kenny J, Fong JC, Takada LT, Ponto C, Hermann P, Knipper T, Stehmann C, Kitamoto T, Ae R, Hamaguchi T, Sanjo N, Tsukamoto T, Mizusawa H, Collins SJ, Chiesa R, Roiter I, de Pedro-Cuesta J, Calero M, Geschwind MD, Yamada M, Nakamura Y, Mead S. Age at onset in genetic prion disease and the design of preventive clinical trials. Neurology. 2019 Jun 6; PMID: 31171647

21. Vallabh SM, Minikel EV, Williams VJ, Carlyle BC, McManus AJ, Wennick CD, Bolling A, Trombetta BA, Urick D, Nobuhara CK, Gerber J, Duddy H, Lachmann I, Stehmann C, Collins SJ, Blennow K, Zetterberg H, Arnold SE. Cerebrospinal fluid and plasma

biomarkers in individuals at risk for genetic prion disease. BMC Med. 2020 Jun 18;18(1):140. PMCID: PMC7302371

22. Vallabh SM, Mortberg MA, Allen SW, Kupferschmid AC, Kivisakk P, Hammerschlag BL, Bolling A, Trombetta BA, Devitte-McKee K, Ford AM, Sather LE, Duffy G, Rivera A, Gerber J, McManus AJ, Minikel EV, Arnold SE. Fluid Biomarkers in Individuals at Risk for Genetic Prion Disease up to Disease Conversion. Neurology. 2024 Jul 23;103(2):e209506. PMCID: PMC11226308

23. Mok TH, Nihat A, Majbour N, Sequeira D, Holm-Mercer L, Coysh T, Darwent L, Batchelor M, Groveman BR, Orrù CD, Hughson AG, Heslegrave A, Laban R, Veleva E, Paterson RW, Keshavan A, Schott J, Swift IJ, Heller C, Rohrer JD, Gerhard A, Butler C, Rowe JB, Masellis M, Chapman M, Lunn MP, Bieschke J, Jackson GS, Zetterberg H, Caughey B, Rudge P, Collinge J, Mead S. Seed amplification and neurodegeneration marker trajectories in individuals at risk of prion disease. Brain J Neurol. 2023 Mar 28;awad101. PMID: 36975162

24. Minikel EV, Vallabh SM. Where have prions been all our lives? Brain J Neurol. 2023 May 10;awad143. PMID: 37161596

25. Fischer M, Rülicke T, Raeber A, Sailer A, Moser M, Oesch B, Brandner S, Aguzzi A, Weissmann C. Prion protein (PrP) with amino-proximal deletions restoring susceptibility of PrP knockout mice to scrapie. EMBO J. 1996 Mar 15;15(6):1255–1264. PMCID: PMC450028

26. Watts JC, Prusiner SB. Mouse models for studying the formation and propagation of prions. J Biol Chem. 2014 Jul 18;289(29):19841–19849. PMCID: PMC4106304

27. Mallucci G, Dickinson A, Linehan J, Klöhn PC, Brandner S, Collinge J. Depleting neuronal PrP in prion infection prevents disease and reverses spongiosis. Science. 2003 Oct 31;302(5646):871–874. PMID: 14593181

28. Safar JG, DeArmond SJ, Kociuba K, Deering C, Didorenko S, Bouzamondo-Bernstein E, Prusiner SB, Tremblay P. Prion clearance in bigenic mice. J Gen Virol. 2005 Oct;86(Pt 10):2913–2923. PMID: 16186247

29. Büeler H, Fischer M, Lang Y, Bluethmann H, Lipp HP, DeArmond SJ, Prusiner SB, Aguet M, Weissmann C. Normal development and behaviour of mice lacking the neuronal cell-surface PrP protein. Nature. 1992 Apr 16;356(6370):577–582. PMID: 1373228

30. Richt JA, Kasinathan P, Hamir AN, Castilla J, Sathiyaseelan T, Vargas F, Sathiyaseelan J, Wu H, Matsushita H, Koster J, Kato S, Ishida I, Soto C, Robl JM, Kuroiwa Y. Production of cattle lacking prion protein. Nat Biotechnol. 2007 Jan;25(1):132–138. PMCID: PMC2813193

31. Yu G, Chen J, Xu Y, Zhu C, Yu H, Liu S, Sha H, Chen J, Xu X, Wu Y, Zhang A, Ma J, Cheng G. Generation of goats lacking prion protein. Mol Reprod Dev. 2009 Jan;76(1):3. PMID: 18951376

32. Benestad SL, Austbø L, Tranulis MA, Espenes A, Olsaker I. Healthy goats naturally devoid of prion protein. Vet Res. 2012;43:87. PMCID: PMC3542104

33. Bremer J, Baumann F, Tiberi C, Wessig C, Fischer H, Schwarz P, Steele AD, Toyka KV, Nave KA, Weis J, Aguzzi A. Axonal prion protein is required for peripheral myelin maintenance. Nat Neurosci. 2010 Mar;13(3):310–318. PMID: 20098419

34. Küffer A, Lakkaraju AKK, Mogha A, Petersen SC, Airich K, Doucerain C, Marpakwar R, Bakirci P, Senatore A, Monnard A, Schiavi C, Nuvolone M, Grosshans B, Hornemann S, Bassilana F, Monk KR, Aguzzi A. The prion protein is an agonistic ligand of the G protein-coupled receptor Adgrg6. Nature. 2016 Aug 25;536(7617):464–468. PMID: 27501152

35. Skedsmo FS, Malachin G, Våge DI, Hammervold MM, Salvesen Ø, Ersdal C, Ranheim B, Stafsnes MH, Bartosova Z, Bruheim P, Jäderlund KH, Matiasek K, Espenes A, Tranulis MA. Demyelinating polyneuropathy in goats lacking prion protein. FASEB J Off Publ Fed Am Soc Exp Biol. 2019 Dec 13; PMID: 31907995

36. Mallucci GR, Ratté S, Asante EA, Linehan J, Gowland I, Jefferys JGR, Collinge J. Post-natal knockout of prion protein alters hippocampal CA1 properties, but does not result in neurodegeneration. EMBO J. 2002 Feb 1;21(3):202–210. PMCID: PMC125833

37. Minikel EV, Karczewski KJ, Martin HC, Cummings BB, Whiffin N, Rhodes D, Alföldi J, Trembath RC, van Heel DA, Daly MJ, Genome Aggregation Database Production Team, Genome Aggregation Database Consortium, Schreiber SL, MacArthur DG. Evaluating drug targets through human loss-of-function genetic variation. Nature. 2020 May;581(7809):459–464. PMCID: PMC7272226

38. Petersen RB, Parchi P, Richardson SL, Urig CB, Gambetti P. Effect of the D178N mutation and the codon 129 polymorphism on the metabolism of the prion protein. J Biol Chem. 1996 May 24;271(21):12661–12668. PMID: 8647879

39. Emami A, Chang JH, Mellon D, Yaksh TL. Local safety assessment of drugs administered via the intrathecal route. Poster. American College of Toxicology 44th Annual Meeting. Int J Toxicol [Internet]. 2023. p. 73–115. Available from: https://doi.org/10.1177/10915818231220242

40. Emami A, Tepper J, Short B, Yaksh TL, Bendele AM, Ramani T, Cisternas AF, Chang JH, Mellon RD. Toxicology Evaluation of Drugs Administered via Uncommon Routes: Intranasal, Intraocular, Intrathecal/Intraspinal, and Intra-Articular. Int J Toxicol. 2018;37(1):4–27. PMCID: PMC5874330

41. Posner K, Oquendo MA, Gould M, Stanley B, Davies M. Columbia Classification Algorithm of Suicide Assessment (C-CASA): classification of suicidal events in the FDA's

pediatric suicidal risk analysis of antidepressants. Am J Psychiatry. 2007 Jul;164(7):1035–1043. PMCID: PMC3804920

42. Thompson AGB, Lowe J, Fox Z, Lukic A, Porter MC, Ford L, Gorham M, Gopalakrishnan GS, Rudge P, Walker AS, Collinge J, Mead S. The Medical Research Council prion disease rating scale: a new outcome measure for prion disease therapeutic trials developed and validated using systematic observational studies. Brain J Neurol. 2013 Apr;136(Pt 4):1116–1127. PMID: 23550114

43. Mead S, Lloyd S, Collinge J. Genetic Factors in Mammalian Prion Diseases. Annu Rev Genet. 2019 03;53:117–147. PMID: 31537104

44. Jones E, Hummerich H, Viré E, Uphill J, Dimitriadis A, Speedy H, Campbell T, Norsworthy P, Quinn L, Whitfield J, Linehan J, Jaunmuktane Z, Brandner S, Jat P, Nihat A, How Mok T, Ahmed P, Collins S, Stehmann C, Sarros S, Kovacs GG, Geschwind MD, Golubjatnikov A, Frontzek K, Budka H, Aguzzi A, Karamujić-Čomić H, van der Lee SJ, Ibrahim-Verbaas CA, van Duijn CM, Sikorska B, Golanska E, Liberski PP, Calero M, Calero O, Sanchez-Juan P, Salas A, Martinón-Torres F, Bouaziz-Amar E, Haïk S, Laplanche JL, Brandel JP, Amouyel P, Lambert JC, Parchi P, Bartoletti-Stella A, Capellari S, Poleggi A, Ladogana A, Pocchiari M, Aneli S, Matullo G, Knight R, Zafar S, Zerr I, Booth S, Coulthart MB, Jansen GH, Glisic K, Blevins J, Gambetti P, Safar J, Appleby B, Collinge J, Mead S. Identification of novel risk loci and causal insights for sporadic Creutzfeldt-Jakob disease: a genome-wide association study. Lancet Neurol. 2020 Oct;19(10):840–848. PMID: 32949544